

TSMaster Release Note

1 Features

1.1 V2026.3.15.1896

1.2 V2026.3.14.1895

1.3 V2026.3.13.1894

1.4 V2026.3.12.1892

1.5 V2026.3.11.1891

[1] Bug fixes until 2026-3-11

1.6 V2026.3.10.1889

[1] Bug fixes until 2026-3-9

1.7 V2026.3.9.1887

1.8 V2026.3.8.1886

1.9 V2026.3.7.1885

1.10 V2026.3.6.1884

[1] Bug fixes until 2026-3-5

1.11 V2026.3.5.1882

[1] New: mini program libraries and forms load order can be managed

[2] New 45 Apis for Flowchart program automation

[3] Bug fixes until 2026-3-4

1.12 V2026.3.4.1879

[1] Bug fixes until 2026-3-3

1.13 V2026.3.3.1878

1.14 V2026.3.2.1877

[1] Bug fixes until 2026-3-1

1.15 V2026.2.28.1874

1.16 V2026.2.27.1872

[1] Bug fixes until 2026-2-26

1.17 V2026.2.26.1870

[1] Bug fixes until 2026-2-25

1.18 V2026.2.25.1868

1.19 V2026.2.24.1867

[1] Bug fixes until 2026-2-24

1.20 V2026.2.12.1855

1.21 V2026.2.11.1854

[1] Bug fixes until 2026-2-10

1.22 V2026.2.10.1852

1.23 V2026.2.9.1851

[1] Bug fixes until 2026-2-9

1.24 V2026.2.7.1849

[1] Bug fixes until 2026-2-6

1.25 V2026.2.6.1847

[1] Bug fixes until 2026-2-5

1.26 V2026.2.5.1845

[1] New MBD components: Lookup table

[2] New option: check performance inside callbacks

[3] Bug fixes until 2026-2-4

1.27 V2026.2.4.1844

1.28 V2026.2.3.1843

[1] Bug fixes until 2026-2-2

1.29 V2026.2.2.1841

[1] Bug fixes until 2026-2-1

1.30 V2026.2.1.1840

1.31 V2026.1.31.1839

1.32 V2026.1.30.1838

1.33 V2026.1.29.1837

[1] Bug fixes until 2026-1-28

1.34 V2026.1.28.1834

1.35 V2026.1.27.1833

[1] Bug fixes until 2026-1-26

1.36 V2026.1.26.1831

[1] Bug fixes until 2026-1-25

1.37 V2026.1.25.1830

[1] Bug fixes until 2026-1-24

1.38 V2026.1.24.1828

1.39 V2026.1.23.1827

[1] Bug fixes until 2026-1-22

1.40 V2026.1.22.1827

[1] Bug fixes until 2026-1-21

1.41 V2026.1.21.1825

1.42 V2026.1.20.1826

[1] Bug fixes until 2026-1-19

1.43 V2026.1.19.1823

[1] Bug fixes until 2026-1-18

1.44 V2026.1.18.1823

1.45 V2026.1.17.1821

[1] Bug fixes until 2026-1-17

1.46 V2026.1.16.1821

[1] Bug fixes until 2026-1-15

1.47 V2026.1.15.1819

1.48 V2026.1.14.1818

[1] Bug fixes until 2026-1-13

1.49 V2026.1.13.1816

1.50 V2026.1.12.1815

[1] Bug fixes until 2026-1-12

1.51 V2026.1.10.1813

1.52 V2026.1.9.1812

[1] Bug fixes until 2026-1-8

1.53 V2026.1.8.1809

[1] New: TOSUN Database Editor added

[2] Bug fixes until 2026-1-7

1.54 V2026.1.7.1808

[1] Bug fixes until 2026-1-6

1.55 V2026.1.6.1807

[1] Bug fixes until 2026-1-5

1.56 V2026.1.5.1806

[1] New Mp and Dll APIs: get hardware id, get_hardware_id_string_upg1, get_hardware_id_array_upg1

[2] New Mp and Dll APIs: get channel mapping alias, get_mapping_property

[3] Bug fixes until 2026-1-4

1.57 V2026.1.4.1805

[1] Bug fixes until 2026-1-3

1.58 V2026.1.3.1804

1.59 V2026.1.2.1803

1.60 V2026.1.1.1802

1.61 V2025.12.31.1801

[1] New: TOSUN Database Editor added

1.62 V2025.12.30.1799

[1] Bug fixes until 2025-12-30

1.63 V2025.12.29.1797

[1] Bug fixes until 2025-12-28

1.64 V2025.12.28.1794

1.65 V2025.12.27.1791

[1] Bug fixes until 2025-12-27

1.66 V2025.12.26.1789

[1] Bug fixes until 2025-12-25

1.67 V2025.12.25.1788

1.68 V2025.12.24.1784

[1] Bug fixes until 2025-12-24

1.69 V2025.12.23.1780

1.70 V2025.12.22.1777

[1] Bug fixes until 2025-12-22

1.71 V2025.12.21.1775

1.72 V2025.12.20.1774

1.73 V2025.12.19.1773

[1] Bug fixes until 2025-12-19

1.74 V2025.12.18.1771

1.75 V2025.12.17.1769

[1] Bug fixes until 2025-12-17

1.76 V2025.12.16.1767

[1] New python api functions: app.ui_show_tsmaster_window, app.
ui_show_hardware_configuration, app. ui_show_channel_mapping_window

[2] Bug fixes until 2025-12-15

1.77 V2025.12.15.1766

[1] Bug fixes until 2025-12-14

1.78 V2025.12.14.1764

1.79 V2025.12.13.1763

1.80 V2025.12.12.1762

1.81 V2025.12.11.1760

[1] Bug fixes until 2025-12-11

1.82 V2025.12.10.1759

[1] New: new struct.cast_to method to bind to destination in TAC

[2] Bug fixes until 2025-12-9

1.83 V2025.12.9.1757

[1] Bug fixes until 2025-12-8

1.84 V2025.12.8.1755

[1] Bug fixes until 2025-12-7

1.85 V2025.12.7.1752

1.86 V2025.12.6.1751

1.87 V2025.12.5.1750

[1] Bug fixes until 2025-12-4

1.88 V2025.12.4.1749

[1] Bug fixes until 2025-12-3

1.89 V2025.12.3.1747

[1] New: Support ISO 13400 in Automated Test module

[2] New: Support QoS configuration in DDS module

[3] Bug fixes until 2025-12-2

1.90 V2025.12.2.1746

1.91 V2025.12.1.1745

1.92 V2025.11.30.1743

1.93 V2025.11.29.1742

[1] New Mp and Dll Api to get system variable type

[2] New Mp and DLL Apis to call system and library function with serialized arguments

[3] Bug fixes until 2025-11-28

1.94 V2025.11.28.1740

[1] [10 Bug fixes until 2025-11-27

1.95 V2025.11.27.1739

[1] [10 Bug fixes until 2025-11-26

1.96 V2025.11.26.1738

[1] [10 Bug fixes until 2025-11-25

1.97 V2025.11.25.1737

[1] - app.metric_get_w_reset_can_frame_interval_stat

[2] [10 Bug fixes until 2025-11-24

1.98 V2025.11.24.1735

[1] New: MP: Add API of metric frame intervals for CAN bus

[2] - app.metric_start

[3] - app.metric_stop

[4] - app.metric_register_can_frame_interval

[5] - app.metric_unregister_can_frame_interval

[6] - app.metric_get_w_reset_frame_interval_stat

[7] [10 Bug fixes until 2025-11-23

1.99 V2025.11.23.1733

1.100 V2025.11.22.1732

1.101 V2025.11.21.1731

[1] New: MP: Add API of metric frame intervals for CAN bus, app.metric_start, app.metric_stop, app.metric_register_can_frame_interval, app.metric_unregister_can_frame_interval, app.metric_get_w_reset_frame_interval_stat

[2] Bug fixes until 2025-11-20

1.102 V2025.11.20.1729

[1] Bug fixes until 2025-11-19

1.103 V2025.11.19.1727

[1] Bug fixes until 2025-11-18

1.104 V2025.11.18.1725

[1] New: Panel PageControl now supports deleting single page

[2] New: System variable direct access supported in C Mini Program

[3] New Option: Generate system var direct access option in mini program, activated by default

[4] New Mp and Dll Apis: get/set element of system var of array type

[5] New: Array typed system variable direct access supported in C Mini Program

[6] New Mp and Dll APIs: generic RBS signal get/set functions

[7] New: Generic RBS Signal direct access supported in C Mini Program

[8] New Option: Generate signal direct access option in mini program, activated by default

[9] New: Device TC1052 supported

[10] Bug fixes until 2025-11-17

1.105 V2025.11.17.1723

1.106 V2025.11.16.1721

1.107 V2025.11.15.1720

1.108 V2025.11.14.1719

[1] Bug fixes until 2025-11-13

1.109 V2025.11.13.1717

[1] Bug fixes until 2025-11-12

1.110 V2025.11.12.1716

[1] Bug fixes until 2025-11-11

1.111 V2025.11.11.1715

[1] Bug fixes until 2025-11-10

1.112 V2025.11.10.1713

[1] New: Highlight to Source/Destination in MBD block diagram

[2] New Demo: Co-Simulation with TSMaster Rpc

[3] Bug fixes until 2025-11-9

1.113 V2025.11.9.1712

1.114 V2025.11.8.1711

1.115 V2025.11.7.1710

[1] Improve: FlexRay Fibex database support v4.1.2

[2] Bug fixes until 2025-11-6

1.116 V2025.11.6.1708

1.117 V2025.11.5.1707

[1] Bug fixes until 2025-11-4

1.118 V2025.11.4.1705

[1] Bug fixes until 2025-11-3

1.119 V2025.11.2.1703

1.120 V2025.11.1.1702

[1] Bug fixes until 2025-10-31

1.121 V2025.10.31.1701

1.122 V2025.10.30.1700

1.123 V2025.10.29.1699

[1] Bug fixes until 2025-10-28

1.124 V2025.10.28.1697

[1] Bug fixes until 2025-10-27

1.125 V2025.10.27.1696

[1] Improved: Repeatedly clicking the Arrange Windows button can swap the window positions

1.126 V2025.10.26.1694

[1] Bug fixes until 2025-10-26

1.127 V2025.10.24.1692

[1] Bug fixes until 2025-10-23

1.128 V2025.10.23.1691

[1] Bug fixes until 2025-10-22

1.129 V2025.10.22.1690

[1] Bug fixes until 2025-10-21

1.130 V2025.10.20.1688

[1] Bug fixes until 2025-10-19

1.131 V2025.10.19.1687

1.132 V2025.10.18.1686

1.133 V2025.10.17.1685

[1] New: Rolling Counter Increment can be set, which allows users to customize the increment value of the Rolling Counter, supporting rapid counting, countdown, test verification, and other scenarios

[2] Bug fixes until 2025-10-16

1.134 V2025.10.16.1684

[1] Bug fixes until 2025-10-15

1.135 V2025.10.15.1683

[1] New Demo: TAC events handler

[2] Bug fixes until 2025-10-14

1.136 V2025.10.14.1682

[1] Bug fixes until 2025-10-13

1.137 V2025.10.13.1680

[1] Bug fixes until 2025-10-12

1.138 V2025.10.12.1677

[1] Bug fixes until 2025-10-11

1.139 V2025.10.11.1676

[1] New Option: Do not show splash window on startup

[2] New Apis: hide and show toolbar of specific window, maximize and restore specific window under normal state

[3] New TAC Events for Each Form: On Configuration Loaded, On Show MDI, On Show Normal, On Show Docked, On Hide, On Destroy

[4] New advanced demo for Signal Tester

[5] Bug fixes until 2025-10-10

1.140 V2025.10.10.1675

[1] Bug fixes until 2025-10-9

1.141 V2025.10.9.1674

[1] Bug fixes until 2025-10-8

1.142 V2025.10.8.1671

[1] Bug fixes until 2025-10-7

1.143 V2025.10.7.1670

1.144 V2025.10.6.1670

[1] Bug fixes until 2025-10-5

1.145 V2025.9.29.1663

1.146 V2025.9.28.1662

[1] Bug fixes until 2025-9-27

1.147 V2025.9.27.1661

[1] Bug fixes until 2025-9-26

1.148 V2025.9.25.1659

[1] Bug fixes until 2025-9-25

1.149 V2025.9.24.1656

[1] TAC now supports installing function with arbitrary arguments as mini program API

[2] New: TAC supports various event sources

[3] Bug fixes until 2025-9-24

1.150 V2025.9.23.1654

[1] New mini program APIs: `app.ui_graphics_set_signal_step_style`,
`app.ui_graphics_set_y_axis_split_mode`

[2] Bug fixes until 2025-9-22

1.151 V2025.9.22.1652

[1] Bug fixes until 2025-9-21

1.152 V2025.9.21.1651

[1] Bug fixes until 2025-9-20

1.153 V2025.9.20.1649

[1] Improved: TAC advanced feature to manipulate system var with namespace, system var with array type, system var in statements

[2] Bug fixes until 2025-9-19

1.154 V2025.9.19.1648

[1] New: Deleted mini program/C Script/Python Code editors can be recovered

[2] New: Drag drop supported: Cpp files as mini programs and C files as C scripts

[3] New: Cpp mini program/C Script/Python Code source code can be removed to recycle bin

[4] New system API: app.ui_graphics_set_split_count to set split count in specific graphics

[5] Bug fixes until 2025-9-18

1.155 V2025.9.18.1646

1.156 V2025.9.17.1644

[1] New: Run history can be replayed in flowchart program

[2] Bug fixes until 2025-9-17

1.157 V2025.9.16.1643

[1] Bug fixes until 2025-9-15

1.158 V2025.9.15.1640

[1] Bug fixes until 2025-9-14

1.159 V2025.9.13.1637

[1] Bug fixes until 2025-9-12

1.160 V2025.9.12.1636

[1] Bug fixes until 2025-9-11

1.161 V2025.9.11.1633

[1] Bug fixes until 2025-9-10

1.162 V2025.9.9.1631

1.163 V2025.9.8.1629

[1] New: Timer group supported in C mini program and Python mini program

[2] New demo project: Timer Group

[3] Bug fixes until 2025-9-8

1.164 V2025.9.7.1627

1.165 V2025.9.6.1626

[1] New: ARINC-825 Database files supported

[2] Bug fixes until 2025-9-5

1.166 V2025.9.5.1624

1.167 V2025.9.4.1623

[1] Bug fixes until 2025-9-3

1.168 V2025.9.3.1621

[1] TAC script can now read write system variables elegantly

[2] New: Run TAC script from string or file in flowchart program

[3] Bug fixes until 2025-9-2

1.169 V2025.9.2.1619

[1] Bug fixes until 2025-9-1

1.170 V2025.9.1.1618

[1] Bug fixes until 2025-8-31

1.171 V2025.8.31.1615

[1] Bug fixes until 2025-8-30

1.172 V2025.8.30.1614

1.173 V2025.8.29.1613

[1] TAC script can now manipulate frames and signals elegantly

[2] Bug fixes until 2025-8-28

1.174 V2025.8.28.1611

[1] Improve: Firmware of TC1055 Pro updated

[2] New mp and dll API for rpc global timeout settings: rpc_set_global_timeout

[3] Bug fixes until 2025-8-27

1.175 V2025.8.27.1607

[1] New: direct execute script API: tac.run_script_sync and tac.run_file_sync

[2] Bug fixes until 2025-8-26

1.176 V2025.8.26.1606

1.177 V2025.8.25.1604

[1] Bug fixes until 2025-8-24

1.178 V2025.8.24.1601

[1] New: MBD diagram now supports exporting ui construction scripts

[2] New: MBD replace block API available

[3] New: global repeat count option in Flowchart program

[4] Bug fixes until 2025-8-23

1.179 V2025.8.23.1600

1.180 V2025.8.22.1599

1.181 V2025.8.21.1598

[1] Bug fixes until 2025-8-20

1.182 V2025.8.20.1596

[1] Bug fixes until 2025-8-19

1.183 V2025.8.19.1595

[1] Bug fixes until 2025-8-18

1.184 V2025.8.18.1594

[1] New Demo Project: TAC Scripting Language Debugger API

[2] Bug fixes until 2025-8-17

1.185 V2025.8.16.1592

1.186 V2025.8.15.1591

1.187 V2025.8.14.1590

[1] Bug fix: code completion proposal font size too small in high-dpi

[2] New: globally disable breakpoint feature added in flowchart program

[3] New Demo Project: TAC Scripting Language Introduction

[4] Bug fixes until 2025-8-14

1.188 V2025.8.13.1588

[1] New: TAC (TOSUN Automation C) scripting language embedded in C Script

[2] New: import MBD C code directly into AiFlow

[3] Bug fixes until 2025-8-12

1.189 V2025.8.12.1586

1.190 V2025.8.11.1583

[1] New Panel Control: Panel Button

[2] Improved: AutoLooupList property added to Selector to auto build select list on runtime

[3] Bug fixes until 2025-8-11

1.191 V2025.8.10.1582

1.192 V2025.8.9.1581

1.193 V2025.8.8.1580

[1] Bug fixes until 2025-8-8

1.194 V2025.8.7.1578

1.195 V2025.8.6.1576

[1] Bug fixes until 2025-8-6

1.196 V2025.8.5.1574

[1] Improve: Firmware of TC1054 Pro updated

[2] Bug fixes until 2025-8-4

1.197 V2025.8.4.1572

[1] Bug fixes until 2025-8-3

1.198 V2025.8.3.1571

1.199 V2025.8.2.1570

1.200 V2025.8.1.1569

[1] New Mp APIs group for model based design

[2] Bug fixes until 2025-7-31

1.201 V2025.7.31.1568

1.202 V2025.7.30.1567

1.203 V2025.7.29.1566

[1] New: Real-time mode in MBD block diagram

[2] New: C Mini program timer now supports start with singleshot mode

[3] New: Python mini program timer now supports start with singleshot mode

[4] New MBD components: Digital Clock, Digital Clock Second, String Constant, Callback Button, Multi Port Switch

[5] Bug fixes until 2025-7-28

1.204 V2025.7.28.1565

1.205 V2025.7.27.1563

1.206 V2025.7.26.1561

1.207 V2025.7.25.1560

[1] Bug fixes until 2025-7-24

1.208 V2025.7.24.1558

[1] Improve: Firmware of TC1055 Pro updated

[2] Bug fixes until 2025-7-23

1.209 V2025.7.23.1555

1.210 V2025.7.22.1554

[1] Bug fixes until 2025-7-21

1.211 V2025.7.21.1552

[1] Bug fixes until 2025-7-20

1.212 V2025.7.20.1549

1.213 V2025.7.19.1548

1.214 V2025.7.18.1547

[1] Bug fixes until 2025-7-18

1.215 V2025.7.17.1544

1.216 V2025.7.16.1539

[1] New: MP API: CAN PDU callbacks API added - can_il_register_autosar_pdu_event, can_il_unregister_autosar_pdu_event, can_il_register_autosar_pdu_pretx_event, can_il_unregister_autosar_pdu_pretx_event

[2] New: MP API: CAN registers configuration API added - configure_can_regs, configure_canfd_regs

[3] Bug fixes until 2025-7-16

1.217 V2025.7.15.1537

1.218 V2025.7.14.1535

[1] Bug fixes until 2025-7-14

1.219 V2025.7.13.1532

1.220 V2025.7.12.1530

1.221 V2025.7.11.1529

[1] Bug fixes until 2025-7-10

1.222 V2025.7.10.1528

[1] New: DDS Module added

[2] Improve: TTS9000 Toolbox refactored

[3] Bug fixes until 2025-7-9

1.223 V2025.7.9.1526

1.224 V2025.7.8.1524

[1] Bug fixes until 2025-7-8

1.225 V2025.7.7.1522

[1] Bug fixes until 2025-7-7

1.226 V2025.7.6.1520

1.227 V2025.7.5.1519

1.228 V2025.7.4.1516

[1] Bug fixes until 2025-7-4

1.229 V2025.7.3.1513

[1] Bug fixes until 2025-7-2

1.230 V2025.7.1.1512

1.231 V2025.6.30.1511

1.232 V2025.6.28.1509

1.233 V2025.6.27.1508

[1] New: A2L Editor added

[2] Bug fixes until 2025-6-26

1.234 V2025.6.26.1505

[1] Bug fixes until 2025-6-25

1.235 V2025.6.25.1503

1.236 V2025.6.24.1502

[1] New: Simulink data dictionary can be imported into MBD data repository

[2] New: search feature added to flowchart program

[3] Bug fixes until 2025-6-23

1.237 V2025.6.23.1499

[1] Bug fixes until 2025-6-22

1.238 V2025.6.22.1498

1.239 V2025.6.21.1497

1.240 V2025.6.20.1495

[1] Bug fixes until 2025-6-20

1.241 V2025.6.19.1493

1.242 V2025.6.18.1491

[1] Bug fixes until 2025-6-18

1.243 V2025.6.17.1489

1.244 V2025.6.16.1488

[1] New: Model based design

[2] New: MBD Manager

[3] New: Variable editor in data repository in mbd module

[4] Bug fixes until 2025-6-16

1.245 V2025.6.15.1485

1.246 V2025.6.14.1484

1.247 V2025.6.13.1483

1.248 V2025.6.12.1480

[1] Bug fixes until 2025-6-11

1.249 V2025.6.11.1478

[1] Bug fixes until 2025-6-10

1.250 V2025.6.10.1475

[1] Bug fixes until 2025-6-9

1.251 V2025.6.9.1473

[1] Bug fixes until 2025-6-8

1.252 V2025.6.8.1471

[1] New: Help: API HTML document added

1.253 V2025.6.7.1470

1.254 V2025.6.6.1469

[1] New: API: MP: show and hide main form

[2] New: User events supported, you can add custom events before or after certain events

[3] Bug fixes until 2025-6-6

1.255 V2025.6.5.1467

1.256 V2025.6.4.1465

[1] New: API: rpc_data_channel_transmit, rpc_data_channel_delete, rpc_data_channel_create added

[2] Bug fixes until 2025-6-4

1.257 V2025.6.3.1461

1.258 V2025.6.2.1458

[1] New: API: lin_rbs_register_force_refresh_frame_by_id added

[2] New: API: lin_rbs_unregister_force_refresh_frame_by_id added

[3] Bug fixes until 2025-6-2

1.259 V2025.6.1.1456

[1] Bug fixes until 2025-5-31

1.260 V2025.5.31.1454

1.261 V2025.5.30.1452

[1] Bug fixes until 2025-5-29

1.262 V2025.5.29.1451

1.263 V2025.5.28.1450

[1] Bug fixes until 2025-5-27

1.264 V2025.5.27.1449

[1] Bug fixes until 2025-5-26

1.265 V2025.5.26.1447

[1] New: API: lin_rbs_update_frame_by_id added

[2] Improved: normal form on top of main form

[3] Bug fixes until 2025-5-25

1.266 V2025.5.23.1446

[1] : New: API: lin_rbs_update_frame_by_id added

[2] Bug fixes until 2025-5-22

1.267 V2025.5.22.1444

1.268 V2025.5.21.1443

1.269 V2025.5.20.1440

[1] Bug fixes until 2025-5-20

1.270 V2025.5.19.1438

[1] Bug fixes until 2025-5-18

1.271 V2025.5.17.1434

1.272 V2025.5.14.1433

[1] Bug fixes until 2025-5-14

1.273 V2025.5.13.1431

1.274 V2025.5.12.1429

[1] Improve: Firmware: TC1054 Pro updated

[2] Bug fixes until 2025-5-12

1.275 V2025.5.9.1426

1.276 V2025.5.8.1425

1.277 V2025.5.7.1424

[1] New: Firmware: TC1055 Pro added

[2] Bug fixes until 2025-5-6

1.278 V2025.5.6.1422

1.279 V2025.5.5.1421

1.280 V2025.5.4.1420

1.281 V2025.5.3.1419

1.282 V2025.5.2.1418

[1] Bug fixes until 2025-5-1

1.283 V2025.5.1.1416

[1] Bug fixes until 2025-4-30

1.284 V2025.4.30.1415

[1] Bug fixes until 2025-4-29

1.285 V2025.4.29.1412

[1] Bug fixes until 2025-4-28

1.286 V2025.4.28.1409

[1] Bug fixes until 2025-4-27

1.287 V2025.4.27.1407

[1] Bug fixes until 2025-4-26

1.288 V2025.4.26.1406

[1] Bug fixes until 2025-4-25

1.289 V2025.4.25.1403

1.290 V2025.4.24.1402

[1] Bug fixes until 2025-4-23

1.291 V2025.4.23.1400

[1] Bug fixes until 2025-4-22

1.292 V2025.4.22.1399

[1] Bug fixes until 2025-4-21

1.293 V2025.4.21.1398

[1] Bug fixes until 2025-4-20

1.294 V2025.4.20.1396

[1] Bug fixes until 2025-4-19

1.295 V2025.4.19.1395

1.296 V2025.4.18.1393

1.297 V2025.4.17.1392

[1] Bug fixes until 2025-4-16

1.298 V2025.4.16.1391

[1] Bug fixes until 2025-4-15

1.299 V2025.4.15.1390

[1] Bug fixes until 2025-4-14

1.300 V2025.4.14.1388

[1] Bug fixes until 2025-4-13

1.301 V2025.4.13.1386

1.302 V2025.4.12.1385

1.303 V2025.4.11.1383

[1] Bug fixes until 2025-4-10

1.304 V2025.4.10.1382

[1] Bug fixes until 2025-4-9

1.305 V2025.4.9.1381

[1] Bug fixes until 2025-4-8

1.306 V2025.4.8.1380

[1] Bug fixes until 2025-4-7

1.307 V2025.4.7.1379

[1] New: drag snapping guideline added for panel controls

[2] Improved: Eat first click before dragging option in panel

[3] Bug fixes until 2025-4-6

1.308 V2025.4.6.1377

1.309 V2025.4.5.1376

1.310 V2025.4.4.1375

1.311 V2025.4.3.1374

[1] Bug fixes until 2025-4-2

1.312 V2025.4.2.1373

[1] Bug fixes until 2025-4-1

1.313 V2025.4.1.1372

[1] Bug fixes until 2025-3-31

1.314 V2025.3.31.1371

[1] Bug fixes until 2025-3-30

1.315 V2025.3.30.1370

1.316 V2025.3.29.1369

1.317 V2025.3.28.1368

[1] Bug fixes until 2025-3-27

1.318 V2025.3.26.1361

1.319 V2025.3.25.1360

1.320 V2025.3.24.1358

[1] Imprvoed: SOME/IP: filter in database view supported

[2] New: Software Gateway for CAN messages added

[3] Bug fixes until 2025-3-23

1.321 V2025.3.23.1357

[1] Bug fixes until 2025-3-22

1.322 V2025.3.22.1355

1.323 V2025.3.21.1354

1.324 V2025.3.20.1353

[1] Bug fixes until 2025-3-19

1.325 V2025.3.19.1352

1.326 V2025.3.17.1350

[1] Bug fixes until 2025-3-16

1.327 V2025.3.10.1348

1.328 V2025.3.9.1347

1.329 V2025.3.8.1346

1.330 V2025.3.7.1345

[1] Bug fixes until 2025-3-6

1.331 V2025.3.6.1344

[1] Imprvoed: GPS data convert from BLF to ASC file format supported

[2] Imprvoed: GPS and ethernet data convert from ASC to BLF file format supported

[3] Bug fixes until 2025-3-5

1.332 V2025.3.5.1342

1.333 V2025.3.4.1341

[1] Bug fixes until 2025-3-4

1.334 V2025.3.3.1339

[1] New: MP: format check for c mini program log functions added by default

[2] Bug fixes until 2025-3-2

1.335 V2025.3.2.1337

1.336 V2025.3.1.1336

1.337 V2025.2.28.1335

[1] Bug fixes until 2025-2-27

1.338 V2025.2.27.1334

[1] Bug fixes until 2025-2-26

1.339 V2025.2.26.1331

[1] Performance improved: massive system variable registration from calibration module

[2] Improved: rebuild system variable tree view with massive system variables

[3] Bug fixes until 2025-2-25

1.340 V2025.2.25.1330

1.341 V2025.2.24.1328

[1] Bug fixes until 2025-2-23

1.342 V2025.2.23.1326

[1] Bug fixes until 2025-2-22

1.343 V2025.2.22.1325

[1] Bug fixes until 2025-2-21

1.344 V2025.2.21.1324

1.345 V2025.2.20.1322

[1] Bug fixes until 2025-2-19

1.346 V2025.2.19.1321

[1] Bug fixes until 2025-2-18

1.347 V2025.2.18.1320

1.348 V2025.2.17.1319

[1] Bug fixes until 2025-2-15

1.349 V2025.2.15.1317

[1] Bug fixes until 2025-2-14

1.350 V2025.2.14.1315

[1] Bug fixes until 2025-2-13

1.351 V2025.2.13.1312

[1] Bug fixes until 2025-2-12

1.352 V2025.2.12.1311

[1] Bug fixes until 2025-2-11

1.353 V2025.2.11.1309

[1] Bug fixes until 2025-2-10

1.354 V2025.2.10.1308

[1] Bug fixes until 2025-2-9

1.355 V2025.2.9.1306

[1] Bug fixes until 2025-2-8

1.356 V2025.2.8.1305

1.357 V2025.2.7.1304

[1] Bug fixes until 2025-2-6

1.358 V2025.2.6.1303

[1] New: Toolbox TIO9000 added

1.359 V2025.2.5.1301

[1] New: Channel mapping in offline replay mode

[2] Bug fixes until 2025-2-5

1.360 V2025.2.4.1299

1.361 V2025.2.3.1298

1.362 V2025.2.2.1297

1.363 V2025.2.1.1296

1.364 V2025.1.31.1295

1.365 V2025.1.30.1294

1.366 V2025.1.29.1293

1.367 V2025.1.28.1292

1.368 V2025.1.27.1291

1.369 V2025.1.26.1290

1.370 V2025.1.25.1289

1.371 V2025.1.24.1288

[1] Bug fixes until 2025-1-23

1.372 V2025.1.23.1286

[1] New: log file order by time

[2] Improved: message counter in Trace window is displayed by channel

[3] New: TSMaster web server can map arbitrary forms as screenshot form

[4] New: TSMaster web server now supports user interactions on screenshot form

[5] Bug fixes until 2025-1-22

1.373 V2025.1.22.1285

1.374 V2025.1.21.1284

1.375 V2025.1.20.1283

1.376 V2025.1.19.1282

[1] New: DBC Editor added

[2] New: TH7011: Delay parameters for transceivers added

[3] New: FMU Generator: Update the Chinese and English user manuals

[4] Bug fixes until 2025-1-19

1.377 V2025.1.18.1280

1.378 V2025.1.17.1278

[1] New: TH7011: Parameters in func "Sample Point Test" added

[2] New: FMU Generator: Update the Chinese and English versions of user manual

1.379 V2025.1.16.1276

1.380 V2025.1.15.1274

1.381 V2025.1.14.1273

[1] Bug fixes until 2025-1-13

1.382 V2025.1.13.1271

[1] New: Mini program library builder allows any dll to be loaded as mini program library

[2] Bug fixes until 2025-1-12

1.383 V2025.1.12.1269

1.384 V2025.1.11.1268

1.385 V2025.1.10.1267

1.386 V2025.1.9.1265

[1] improved: system var separator now supports "::"

[2] New mp and dll API: TSMaster Rpc Ip trigger data group

[3] Bug fixes until 2025-1-9

1.387 V2025.1.8.1261

[1] TSMaster Rpc Ip now supports data groups automatic reporting

[2] Bug fixes until 2025-1-8

1.388 V2025.1.7.1256

[1] New mp and dll APIs: clear and set update bits for CAN and FlexRay

[2] Improved: Release mode supported in panel push button

[3] New mp and dll APIs: generic USB device insertion and removal events handle in mini program

[4] Bug fixes until 2025-1-7

1.389 V2025.1.6.1252

[1] Improved: Real-time co-simulation now supports starting simulation after application is connected

[2] New: Update Bits auto handle supported in CAN AutoSAR database

[3] New mp and dll APIs: clear update bits for CAN and FlexRay

[4] Bug fixes until 2025-1-6

1.390 V2025.1.5.1251

[1] AutoSAR E2E CAN protection and verification fully supported for Profile 01, 02, 04, 05, 06, 07, 11 and 22

[2] New Mp and Dll APIs: CAN RBS fault inject apis for AutoSAR E2E CRC and RC calculation

[3] Bug fixes until 2025-1-5

1.391 V2025.1.4.1250

[1] AutoSAR E2E protection and verification fully supported for Profile 01, 02, 04, 05, 06, 07, 11 and 22

[2] Bug fixes until 2025-1-3

1.392 V2025.1.3.1247

[1] New: Import *.vsysvar files to System Variable added

[2] New Api in Matlab Controller: run_matlab_cmd

[3] New: AutoSAR E2E Viewer

[4] Bug fixes until 2025-1-2

1.393 V2025.1.2.1247

[1] New: Option to install API to ECUTest added

[2] Bug fixes until 2025-1-1

1.394 V2025.1.1.1245

1.395 V2024.12.31.1245

[1] Improved: Timer in C mini program now supports 0.5 ms interval and 0 ms interval (single shot mode)

[2] Bug fixes until 2024-12-30

1.396 V2024.12.30.1243

[1] improved: system variable manager has tree view and flat viewing options

[2] Bug fixes until 2024-12-29

1.397 V2024.12.28.1243

1.398 V2024.12.27.1243

1.399 V2024.12.26.1242

[1] New Panel component: Animated Image supporting gif, webp, tgs, lottie animated images

[2] improved: default system var separator changed to ::

[3] Bug fixes until 2024-12-25

1.400 V2024.12.25.1241

1.401 V2024.12.24.1244

1.402 V2024.12.23.1240

[1] Bug fixes until 2024-12-23

1.403 V2024.12.22.1239

[1] Improved: Panel interface does not require rebuild after form style change

[2] Improved: Value table display in Graphics

[3] New: Increment instance feature in mini program library

[4] Bug fixes until 2024-12-22

1.404 V2024.12.18.1238

[1] New: API: fifo APIs added in MP for bus messages

[2] Bug fixes until 2024-12-18

1.405 V2024.12.17.1235

[1] New: Docking Form Style

[2] New: E2E: CRC8 for Profile 11 added

[3] Bug fixes until 2024-12-17

1.406 V2024.12.16.1233

[1] Bug fixes until 2024-12-16

1.407 V2024.12.15.1231

[1] New: Different background color used based on the theme

[2] Bug fixes until 2024-12-15

1.408 V2024.12.13.1230

[1] Bug fixes until 2024-12-12

1.409 V2024.12.9.1228

[1] New: Lock UI feature implemented for each form in case the form is accidentally modified

1.410 V2024.12.6.1228

1.411 V2024.12.5.1227

[1] Improved: Panel graphics now supports setting color and y min max

[2] New project option to show form by switching page so as to maintain existing form layout within page

[3] New: stop loading and terminate feature added to splash window

[4] New: API: `tssocket_add_ipv4_device` and `tssocket_delete_ipv4_device` added

[5] Bug fixes until 2024-12-04

1.412 V2024.12.3.1226

[1] New: TM2101 (GPS2CAN) functionality added

[2] Bug fixes until 2024-12-02

1.413 V2024.12.2.1225

[1] New: DPI awareness settings in system settings

[2] Bug fixes until 2024-12-01

1.414 V2024.11.29.1224

[1] Improved: block width made wider in graphic program

[2] Improved: Numeric window now supports editing system variable directly

[3] New: Bus logger can be started or stopped by changing corresponding system variable

[4] New: Graphics data points can be exported to CSV file for post processing

[5] New mini program API to export graphics data points to CSV

[6] Bug fixes until 2024-11-28

1.415 V2024.11.26.1223

[1] Bug fixes until 2024-11-25

1.416 V2024.11.25.1222

[1] Bug fixes until 2024-11-24

1.417 V2024.11.20.1218

[1] New: excel test system now supports [CAN], [LIN], [FlexRay], [Sys], [Var], [Const], [File] and [Path] prefix in function argument

[2] New: excel test system now supports system variable argument as return value in api call

[3] New: excel test system can call arbitrary system APIs and library APIs

[4] Bug fixes until 2024-11-20

1.418 V2024.11.18.1217

1.419 V2024.11.17.1216

[1] New mini program and dll API for CAN, LIN, FlexRay RBS start and stop commands

[2] New: CAN RBS controller implemented in TSMaster Web Server

[3] New: FMU Generator added

[4] Bug fixes until 2024-11-16

1.420 V2024.11.16.1216

[1] Bug fixes until 2024-11-15

1.421 V2024.11.15.1215

[1] Bug fix for graphics not displaying occasional maximum and minimum values

[2] New events for toolbox TSTreeList component: node check changing and changed

[3] Bug fixes until 2024-11-14

1.422 V2024.11.13.1214

[1] New mini program API: cal_clear_measurement_items

[2] Bug fixes until 2024-11-12

1.423 V2024.11.11.1213

[1] Bug fixes until 2024-11-10

1.424 V2024.11.10.1213

1.425 V2024.11.8.1212

[1] Bug fixes until 2024-11-7

1.426 V2024.11.7.1211

[1] New API to set graphics y axis range min and max under fixed mode

[2] Improved: real-time comment can be displayed in multilines

[3] New: python can load any plugin and run standalone

[4] New mini program API to export system messages to external file

[5] New: real-time comments are now rendered in graphics with each signal's y value

[6] Bug fixes until 2024-11-6

1.427 V2024.11.5.1210

[1] New mini program API: you can import TPnl configuration file into any existing panel

[2] New mini program and dll API for logging system variables into blf:

tslog_blf_write_sysvar_double

tslog_blf_write_sysvar_s32

tslog_blf_write_sysvar_u32

tslog_blf_write_sysvar_s64

tslog_blf_write_sysvar_u64

tslog_blf_write_sysvar_string

tslog_blf_write_sysvar_double_array

tslog_blf_write_sysvar_s32_array

tslog_blf_write_sysvar_u8_array

[3] Bug fixes until 2024-11-4

1.428 V2024.11.4.1209

[1] Bug fixes until 2024-11-3

1.429 V2024.11.3.1208

1.430 V2024.11.1.1207

[1] New: Use colorful themes

1.431 V2024.10.31.1206

[1] New Panel Control: File Selector for user to select and process files automatically

[2] New TSMaster Web Server Component: File Uploader allowing user to upload files remotely to TSMaster server and trigger file processing events

[3] Bug fixes until 2024-10-30

1.432 V2024.10.30.1204

[1] New: VSCode can be automatically configured for python environment

[2] New: you can insert your own UI design into any existing TS Form using python toolbox

[3] New mini program API: `app.write_realtime_comment_w_time` to specify event with precise timestamp

[4] New: Set relative time allows user to analyze signal relationships with specific time offset

[5] New Mini Program API: `app.ui_graphics_set_relative_time` to automatically set relative time in graphics for analysis automation

[6] New: TS Edit List control added in python toolbox allowing cells to be edited

1.433 V2024.10.25.1202

[1] Bug fixes until 2024-10-25

1.434 V2024.10.23.1200

[1] Bug fixes until 2024-10-23

1.435 V2024.10.21.1200

[1] New: Toolbox Test System Lite added, which organizes test cases

[2] New: Toolbox TSMaster RPC Editor added

[3] New: Bypass for TC1054 Pro added

[4] Bug fixes until 2024-10-20

1.436 V2024.10.20.1198

[1] Firmware of TC1054 Pro updated

[2] Bug fixes until 2024-10-19

1.437 V2024.10.14.1198

[1] Bug fixes until 2024-10-13

1.438 V2024.10.12.1198

1.439 V2024.10.11.1198

1.440 V2024.10.10.1197

[1] Bug fixes until 2024-10-10

1.441 V2024.10.9.1196

[1] New: Toolbox TH7011 for CAN disturbance added

[2] New: Toolbox TTS9000 for TOSUN Card added

[3] Bug fixes until 2024-10-09

1.442 V2024.10.7.1194

[1] Bug fixes until 2024-10-06

1.443 V2024.10.6.1194

1.444 V2024.10.2.1193

1.445 V2024.10.1.1192

[1] Bug fixes until 2024-09-30

1.446 V2024.9.29.1190

[1] Doc: Quick start in English added

[2] Bug fixes until 2024-09-28

1.447 V2024.9.28.1189

1.448 V2024.9.27.1188

1.449 V2024.9.26.1187

[1] New Form: Graphic Program Library for program modularization

[2] New: Graphic program sub module can be moved to another graphic program

[3] t7z, pt7z and tsapp file types now follows project selector

[4] New: Graphics now supports range statistics calculation between diff cursors

[5] Improved: column best fit feature in Graphics

[6] Bug fixes until 2024-09-26

1.450 V2024.9.24.1186

1.451 V2024.9.23.1185

[1] New: Treelist component in Python Toolbox

[2] New mini program Apis: app.ui_load_rpc_ip_configuration,
app.ui_unload_rpc_ip_configuration and app.ui_unload_rpc_ip_configurations

[3] New: User can add custom columns for additional signal display in graphic program

[4] New mini program API: auto add custom columns into graphic program

[5] Bug fixes until 2024-09-22

1.452 V2024.9.19.1184

[1] New Python Component: TSSplitter

[2] New: mini program APIs can be imported as header and called freely from toolbox

[3] New: mini program APIs can be imported as header and called freely from python mini program

1.453 V2024.9.18.1184

[1] Bug fixes until 2024-09-18

1.454 V2024.9.17.1183

[1] Improved: One of muptile A2ML definitions in an A2L file can be selected

[2] Bug fixes until 2024-09-16

1.455 V2024.9.14.1182

[1] New Mp API: test.log_formatted_value

[2] New TSMaster Rpc Ip Request Command: Retrieve xml = 0x12; Response Command:
Retrieve xml = 0x13

[3] New python APIs: log file converter functions for blf, csv, asc and mat files

[4] New Module: Java Environment, Java APIs can be called natively

[5] New: Mini program API library builder for Java. TSMaster mini program / graphic program now call Java freely

[6] New: Graphic program builder for Java API

[7] New Demo: Java API Caller

[8] Bug fixes until 2024-09-13

1.456 V2024.9.13.1182

1.457 V2024.9.11.1182

1.458 V2024.9.10.1181

[1] Bug fixes until 2024-09-09

1.459 V2024.9.9.1180

[1] Bug fixes until 2024-09-08

1.460 V2024.9.5.1179

[1] new mp api: test.log_formatted_value, com.cal_get_data_by_row_and_col, com.cal_set_data_by_row_and_col

[2] Improved: Symbol mapping filter direction can be set

[3] Bug fixes until 2024-09-04

1.461 V2024.9.4.1178

1.462 V2024.9.2.1178

[1] Bug fixes until 2024-09-01

1.463 V2024.8.31.1177

[1] Bug fixes until 2024-08-31

1.464 V2024.8.29.1176

[1] New: FMI Controller supports loading FMU version 1.0, 2.0 and 3.0

[2] New: Auto export asc file supported in logging module

[3] Bug fixes until 2024-08-28

1.465 V2024.8.27.1176

[1] Bug fixes until 2024-08-26

1.466 V2024.8.26.1174

[1] Bug fixes until 2024-08-25

1.467 V2024.8.24.1173

1.468 V2024.8.23.1172

1.469 V2024.8.22.1171

[1] New: Device TO1013 added

[2] New: Device TC1054 Pro added

[3] Bug fixes until 2024-08-22

1.470 V2024.8.21.1170

[1] Bug fixes until 2024-08-20

1.471 V2024.8.20.1168

1.472 V2024.8.19.1168

[1] New: LDF Editor added

[2] Bug fixes until 2024-08-19

1.473 V2024.8.17.1166

[1] New API: ui_save_project

[2] Bug fixes until 2024-08-16

1.474 V2024.8.16.1165

[1] Bug fixes until 2024-08-15

1.475 V2024.8.15.1164

[1] New: Instrument DG10XX added

[2] New: Global control feature added for user control over simulation start/stop

[3] New Panel API: create control, delete control, get control and set control

[4] Bug fixes until 2024-08-14

1.476 V2024.8.12.1163

[1] New: Mini program API visibility can be changed for project deployment

1.477 V2024.8.7.1162

[1] Bug fixes until 2024-08-06

1.478 V2024.8.6.1161

1.479 V2024.8.5.1159

[1] New: Multiple TSMaster with different versions instances can be deployed on same computer

[2] New: simulation performance evaluation added in Rpc module

[3] Grid is displayed for panel design

[4] New: Map Viewer functionality added, which supports TOSUN log devices

[5] New: Instruments IT63XX, IT65XX, Agilent34461 added

[6] Bug fixes until 2024-08-05

1.480 2024-07-30

[1] Improved: Python toolbox display name can be changed

1.481 2024-07-26

[1] New: Singleton mode for python toolbox window

[2] New API: auto create and delete python toolbox icon on ribbon

1.482 2024-07-25

[1] new api: is_realtime_mode and is_simulation_mode

[2] New: user permission control in TSMaster web server

[3] New: TSMaster version selector to control default project opening behavior

[4] New: TSMaster version selector to control specific project opening behavior

[5] New: Lock encrypt published project with specific TSMaster version

[6] New: Lock app with specific TSMaster version

1.483 2024-07-21

[1] New: Multiple TSMaster with different versions instances can be deployed on same computer

1.484 2024-07-19

[1] Bug fixes until 2024-07-19

[2] New: TSMaster Web Server released for remote calibration, remote diagnostics, remote monitoring with Panels

[3] New: TSMaster Simulink Package released for model in the loop (MIL) cosimulation

[4] New API: com.can_rbs_set_crc_signal_w_head_tail

1.485 2024-07-17

[1] Bug fixes until 2024-07-17

1.486 2024-07-16

[1] New: RP1210-C functionality added

1.487 2024-07-11

[1] New: Online Documents entry on help ribbon

1.488 2024-07-10

[1] New APIs: signal server cache registration and value retrieve features in TSMaster Rpc

[2] Improved: mini program build system is used by default

[3] New: TSMaster Rpc Ip configuration file can be converted to Cpp and Python projects

1.489 2024-07-04

[1] bug fixes until 2024-07-04

[2] classic test sytem can be export w/wo source

1.490 2024-06-30

[1] Bug fixes until 2024-06-30

1.491 2024-06-28

[1] Bug fixes until 2024-06-28

1.492 2024-06-27

[1] New: Python code editor can now remember database symbol settings [2] New C and Python API: encode and decode string with base64

1.493 2024-06-26

[1] Bug fixes until 2024-06-26

[2] New: C code editor can now remember database symbol settings

1.494 2024-06-23

[1] Bug fixes until 2024-06-23

1.495 2024-06-20

[1] Improved: Excel Parsing Engine Update: Performance Triples

1.496 2024-06-19

[1] New Python toolbox control: ListView

1.497 2024-06-17

[1] New Mp APIs for cursor locating in Graphics

1.498 2024-06-05

[1] New: IXXAT devices supported

[2] New C and Python API: get form instance count

[3] New C and Python API: app.get_active_application_list

1.499 2024-06-01

[1] Improved: Mini program build speed optimized

1.500 2024-05-31

[1] New APIs: mini program library control in mini program

[2] New APIs for C and Python:

[3] app.show_splash_message(AMsg: str): int

[4] app.tsmp_get_configuration_address(AMpFileName: str): int

[5] app.get_tsmaster_binary_location(): str

[6] test.classic_test_system_import(AConfFile: str): int

[7] test.classic_test_system_login(AUserName: str, APassword: str): int

[8] Improved: RBS Signal address can be simplified if not named identically

1.501 2024-05-29

[1] New: Generic output function for all kinds of bus messages transmission

[2] Improved: compiling C mini program will not block whole application UI

[3] Improved: Parallel compilation of all auto start mini programs before application connection supported

[4] New: TSMaster Rpc Ip released. Compatible with CANoe's FDX (Fast Data Exchange) protocol

1.502 2024-05-24

[1] Bug fix: large font in high DPI screen

1.503 2024-05-22

[1] New: Last value column added in graphics program for signal read/write, API call and arbitrary expression

[2] New: Multiplexed signal now supports value table range

1.504 2024-05-16

[1] New: TSMaster Rpc Co-simulation APIs added for arbitrary signal read and write and arbitrary API function call in remote TSMaster

1.505 2024-05-15

[1] New: System variable value table can be exported as c header file

1.506 2024-05-09

[1] Recompile All now supports: C mini program, test cases and graphic programs

[2] New: Restrict jump inside current action group in graphic program

[3] New APIs for system variables init value control

[4] New: watermark supported in graphic program reporting

[5] New: Action UI positions can be locked or unlocked in graphic program

1.507 2024-04-30

[1] Arbitrary signal manipulation expression supported in CAN RBS, writing preTx is not needed

[2] Arbitrary signal expression editor added

1.508 2024-04-27

[1] Improved: drag drop components in toolbox form designer supported

1.509 2024-04-24

[1] New: Parallel compile feature to improve mini program compile speed

[2] New: TSMaster PCH file option is activated as default to speed up mini program compile speed

[3] New APIs to retrieve mini program info and compile mini programs

1.510 2024-04-18

[1] New APIs: user constants can be loaded from c header file

[2] New: Bypass pretx in CAN, LIN, FlexRay transmit window

1.511 2024-04-15

[1] New: APIs for co-simulation added

1.512 2024-04-07

[1] New: full simulated mode supported

1.513 2024-03-30

[1] New: TSMaster Rpc Server and Client demo added

1.514 2024-03-29

[1] Upgraded: mini program now supports clang 18

[2] New: TSMaster Rpc Server and Client are now available

[3] New: TSMaster Rpc APIs added in mini program and dll

1.515 2024-03-28

[1] New: TSMaster toolbox manual released

1.516 2024-03-26

[1] Generic Rpc Server and Client API added in mini program and dll

[2] New group for co-simulation between different simulation software

[3] New API to mask FPU exceptions such as divide by 0 in simulation environment

1.517 2024-03-21

[1] Bug fix: dpi scale in different monitors

1.518 2024-03-16

[1] Bug fix: utf-8 string supported in mini program function arguments

1.519 2024-03-08

[1] TSMaster overseas version released

1.520 2024-03-06

[1] New Toolbox API for system variable change callback support

1.521 2024-03-05

[1] New API to compensate timestamp deviation in multiple vendor hardware

[2] New API: output system message log to a directory

1.522 2024-02-27

[1] New APIs: loading and unloading databases supported in mini program

1.523 2024-02-26

[1] New mini program API for critical section

[2] New: Constant manager now can parse C header file

[3] Bug fixes for UI

1.524 2024-02-23

[1] TSMaster x64 is out

[2] Mini Program now supports Clang 17

1.525 2024-02-07

[1] New toolbox released: GW2112 configurator

[2] New toolbox released: Code package generator

1.526 2024-02-06

[1] New: Pcapng ethernet log file supported in Replay

[2] New: Pcapng ethernet log converting supported in converter

[3] New mini program and dll APIs: pcapng to blf and blf to pcapng

1.527 2024-02-05

[1] Bug fix: automation module verdict color

1.528 2024-02-04

[1] New: Pcap ethernet log file supported Replay

[2] New: Pcap ethernet log converting supported in converter

[3] New mini program and dll APIs: pcap to blf and blf to pcap

1.529 2024-01-31

[1] Improved: High DPI support

[2] Improved: UI performance

1.530 2024-01-30

[1] Improved: Toolbox UI designer now supports dragging multiple selected components

[2] New: Toolbox UI now supports non TSMaster Form design

[3] New: Telnet APIs added in mini program and dll

[4] New Demo: Telnet Communication

1.531 2024-01-26

[1] New Python APIs for Component and Control iteration

1.532 2024-01-25

[1] Improved: Graphics grid now follows axis markers

1.533 2024-01-24

[1] New: Graphics signal now supports arbitrary expressions

[2] New: create_form API added for UI automation

1.534 2024-01-23

[1] Bug fix: CAN Trace distinguish same identifier with standard and extended

1.535 2024-01-22

[1] New APIs for form operation: show, hide, toggle MDI, set positions

[2] New Toolbox: GW2112 for TOSUN gateway devices

[3] New: Drag drop supported for system variables config file

1.536 2024-01-19

[1] Refactored: Graphics now accurately displays sporadic abnormal signal points within a vast number of signal points

[2] New: Panel API supports clear any panel control contents

1.537 2024-01-17

[1] New: Panel TPnl files can be directly dragged into TSMaster Interface

[2] Improved: NOK results in NOP now generates error info in Graphic Program

1.538 2024-01-12

[1] Improved: Window flickers when switch between MDI and Normal State

[2] Improved: cyclic message added by API become more precise

1.539 2024-01-11

[1] New: Ethernet blf asc converter APIs available

1.540 2024-01-08

[1] Bug fix: main form position changed on monitors with different DPIs

1.541 2024-01-06

[1] New: UDP fragment processor API added

[2] New: VLAN APIs added: append, insert, remove

[3] New: E2E supports classic CAN frames

[4] New: Bus Replay Post processing now supports LIN, FlexRay and Ethernet

[5] New: Generic Plugin now supports multi-language

1.542 2023-12-27

[1] New: Generic Cpp demo plugin added to extend TSMaster Capabilities in C++

1.543 2023-12-22

[1] New: Cpp demo plugin added to extend TSMaster Capabilities in C++

1.544 2023-12-19

[1] New: your python toolbox source code can be compiled to pyd for software protection

1.545 2023-12-18

[1] New: Mini program library exporting support added in toolbox for auto installation when being imported by another TSMaster

[2] New: Option for Auto updating FlexRay frame in RBS

1.546 2023-12-16

[1] New: FlexRay E2E CRC and RC supported in RBS

1.547 2023-12-15

[1] New: Add new Device TP1026

[2] New: storing and reading calibration values through Excel

[3] Bug fix: When calibrating sub elements of the same Map/Curve Var, the number of operations should not be limited

1.548 2023-12-14

[1] New: Standalone Python IDE for General Application Development

[2] New: Style can be selected in Toolbox UI designer

1.549 2023-12-12**1.550 2023-12-10**

[1] New: mat file converter in logging and replay window

[2] New APIs for log file converting: blf, asc, csv, mat

[3] Log Converter refactor

[4] Minor bug fixes

[5] Max supported channel count set to 64

[6] New API: get form unique id for caption search

[7] New API: set debug log level in mp and dll

- [8] New Python API: `app.ui_get_component`, everything in TSMaster can be manipulated !
- [9] New Python APIs for mini program library operation
- [10] New: Custom Help file can be specified w/wo language support in Panel and Toolbox
- [11] New python API: `call_system_api` and `call_library_api`
- [12] Improved: Filtering in CAN RBS and FlexRay RBS speed up 100x+
- [13] Improved: Toolbox High DPI Aware
- [14] New: Dynamic language supported in toolbox
- [15] Improved: Graphic Program can import and export unlimited nested subroutines
- [16] Improved: Multi-thread supported in Toolbox

1.551 2023-11-25

- [1] New: Multi-language supported in Panel
- [2] Improved: Panel Basics Demo now supports multi-language
- [3] New API: `app.get_language_string` for generic language translation in all your programs
- [4] New: Toolbox UI now supports multi-language
- [5] New: Toolbox treeview now supports more events and methods
- [6] New: Built-in Blf to CSV converter introduced with filter configuration dialog
- [7] New: 2 APIs added for auto CSV file conversion
- [8] New API: `set_flexray_ub_bit_auto_handle`
- [9] New APIs: signal tester get status and set dynamic time and value range
- [10] New: Graphics can now monitor signal's cycle time
- [11] New: online replay now supports code generation
- [12] New: Python Pillow Package example added in toolbox
- [13] New: CAN FD E2E Checksum and Rolling Counter auto detect and handler supported
- [14] New: Python package updated for matrix calculation performance
- [15] New API: `app.start_log_w_filename`
- [16] New: TSMaster toolbox can import and export bundle

- [17] New: Calibration variables can be logged into blf and replayed from blf
- [18] New: Set the A2L database for calibrating ECU by system variables
- [19] New: Splitting an array data(VAL_BLK) into multiple individual variables
- [20] Bug fixes for multiple components

1.552 2023-10-27

- [1] New: Configurations in backup folder can be recovered
- [2] New: Graphic program can now export to Excel
- [3] New Toolbox Control: Check List Box
- [4] New: Toolbox Paint Box control can save PNG file to disk
- [5] New API: test.set_classic_test_system_report_name
- [6] New API: run procedure in main UI thread
- [7] Improved: Signal tester automation and result retrieve
- [8] New: License Protection supported in Encrypt Publish and App Publish
- [9] New: Globally unique Hardware ID can be retrieved
- [10] New: Current project license status information window
- [11] Improved: Project startup information updated with license information
- [12] New: Project License Manager
- [13] New Demo: Popup menu usage in Python toolbox
- [14] New: Standalone toolbox application can be deployed independent of TSMaster
- [15] New: Standalone toolbox application can be executed independent of TSMaster
- [16] New API: show help file and navigate to specified chapter

1.553 2023-10-10

- [1] New API: Graphics signals can be dynamically added by scripts
- [2] Bug fix: Classic Test System string type parameters
- [3] Bug fix: Meter control display and popup menu
- [4] Improved: Mini program documents
- [5] Bug fix: Logging of ethernet frame

[6] New: C code Editor now switches to single cpp file solution, externally editing your mini program code is possible

[7] New API: auto deploy graphic program from excel

[8] Update: Optimize file loading solution to short the loading time.

[9] New: DAQ Timestamp for XCP

[10] New: custom function documentation can be freely edited in C code editor

[11] Improved: Manually add new functions is supported in external C code editor source file

[12] New: Your mini program library can now generate documents automatically for your customers

[13] Bug fix: event properties in mini program lost

1.554 2023-09-22

[1] New: Intelligent C Code Completion feature supported

[2] New: Intelligent C Code Params feature supported

[3] New APIs for Python for path operation in environment

[4] New toolbox component: TreeView

[5] New: toolbox form can be placed anywhere on TSMaster ribbon

[6] New: Symbol Value Window

[7] New: modify signal value from Graphics

[8] New: FlexRay Arxml database supported

[9] Improved: Meter toolbox refactor

[10] Improved: A2L Parser refactor

1.555 2023-09-05

[1] New: Global switch for database header generation for C Code

[2] New: Ethernet Trace Window added

[3] New Devices: TE1051, TP1018, TC1017

[4] Bug fixes for RBS, Panel and more

[5] Python x64 environment can be automatically deployed

[6] New Graphics Options for floating cursors display and Y axis width

[7] New Feature: Signal Tester in Test Solution

[8] New: Alias name supported in Graphics

[9] Bug fix: TSMaster dll init error

1.556 2023-08-04

[1] Improved: tsapp_show_tsmaster_window can now direct start

[2] New: Ethernet support added in mini program

[3] New: TS Virtual Channel and VN5620 series supported in TSMaster with raw ethernet frame Tx/Rx

[4] New APIs: get database index by id

[5] New APIs: ethernet transmission and blf logging

[6] New API for IoIp: com.ioip_set_tcp_server_connection_callback

[7] Improved: all old style open and save dialog upgraded to new style

[8] New APIs: register and unregister system variable change event in mp and dll

[9] Bug fix: Python in TSMaster sometimes cannot use pip

[10] New API: block_current_pretx to block frame transmission in preTx event gracefully

[11] New Python API: flexray_rbs_update_frame_by_header

[12] Bug fixes for UI

[13] Improved: load and save configuration

[14] New Device support: OBD Wifi Device

[15] New: Alias support for CAN and FlexRay database network name, changing database will not affect one line of your code now

[16] New: Arbitrary function caller API for MP, dll and COM, your custom API functions can be invoked dynamically

[17] Improved: Logging start timestamp aligned to application connection time

[18] New: programmable filter for each measurement window, you can write your own code to filter measurement frames

[19] Improved: Graphics lower CPU consumption

1.557 2023-07-11

- [1] New APIs for UI, watchdog and graphics configuration dynamic loading
- [2] New instant API document help in C code editor (F1 is hot key)
- [3] New instant API document in graphic program
- [4] New: Python file in toolbox builder can now be encrypted for distribution
- [5] New: Embed in group command in Graphic program
- [6] New: LIN RBS APIs available in mini program and DLL functions
- [7] New Device: TP1013 supported
- [8] New API for path adding and deleting
- [9] J2534 released
- [10] Improved: TSMaster can now be installed into non-ANSI directory
- [11] Feedback entry added in each Form
- [12] Bug fix: Panel Memo flicker
- [13] New 22 APIs for system var written with timestamp
- [14] New API: app.db_get_signal_startbit_by_pdu_offset
- [15] New: Alignment feature in Toolbox designer
- [16] New APIs for ui operations: ui_show_save_file_dialog, ui_show_open_file_dialog, ui_show_select_directory_dialog
- [17] Improved: channel selection for specific serial number can now be remembered to correct hardware index

1.558 2023-06-14

- [1] New Graphic program importer from Excel file
- [2] New option to improve mini program compile speed (default ON)
- [3] New: CDM module in Calibration

1.559 2023-06-02

- [1] New 32 MP and DLL APIs for Graphic program generation
- [2] New Demo project for Graphic program generation
- [3] Bug fix for mini program build with conflicted define

[4] Improved: mini program OnTx and OnRx data protected by default

[5] New MP API to trigger FlexRay RBS update frame:
`com.flexray_rbs_update_frame_by_header`

1.560 2023-06-01

[1] New MP and DLL API for CAN, LIN and FlexRay raw signal read and write

[2] New option in C Code Editor for generation of raw signal read and write interface

1.561 2023-05-29

[1] New MP and DLL API for FlexRay database: `db_get_flexray_cluster_parameters` and `db_get_flexray_controller_parameters`

[2] New MP and DLL API for System Vars: `set_system_var_event_support` and `get_system_var_event_support`

[3] New MP and DLL API: `get_date_time`

[4] New MP and DLL API for online replay: `tslog_disable_online_replay_filter` and `tslog_set_online_replay_filter`

[5] Signal mapping engine improvement: auto activate RBS, and auto set direction

1.562 2023-05-24

[1] All 450+ mini program APIs exported for Python, demos provided for each API

[2] Auto save project feature added

[3] New api: add and delete system constant

[4] New panel control: Scroll Box hosting sub controls with scroll bars

[5] New toolbox control: Scroll Box hosting sub controls with scroll bars

[6] New converter: FlexRay asc to blf, FlexRay blf to asc

[7] TsFlash device published

[8] New option in App Publish: run with TOSUN hardware w/wo Encrypt publish license

1.563 2023-04-27

[1] New Options for easy signal selection in panel control with edit abilities

[2] New: direct signal mapping now supports CAN, LIN and FlexRay database selectors

[3] New: Static Heading and Trailing bytes can be added to CRC algorithms in CAN and FlexRay RBS

- [4] Improved: Panel Selector now supports FlexRay signal value table
- [5] Improved: Rolling counter increment only available when UB = 1 in FlexRay
- [6] Exported: tsapp_get_flexray_channel_count, tsapp_set_flexray_channel_count
- [7] Exported: generic signal read write APIs for CAN, LIN and FlexRay
- [8] New Option: result can be ignored in graphic program API call
- [9] Improved: Left Jump supported in Graphic Program (Simple support for If statement)

1.564 2023-04-10

- [1] New APIs: get signal properties by index in frame
- [2] New Python APIs: full database parser for CAN, LIN and FlexRay
- [3] New Python APIs: rbs functions for CAN and FlexRay
- [4] New Python APIs: signal get set functions for CAN, LIN and FlexRay
- [5] New demo: Python API tests for all available python functions, increasing
- [6] New large Icons in ribbon
- [7] New: Python mini program now supports database header and constant header
- [8] New: python form now supports id filters for frame reception, transmission and preTx events

1.565 2023-03-29

- [1] New: Configuration manager in graphics
- [2] Bug fix: signal server for fibex signals
- [3] Bug fix: toolbox builder Glyph properties
- [4] New APIs: frame list and signal list can be retrieved from CAN, LIN, FlexRay database in MP and Win32

1.566 2023-03-24

- [1] New: Memo control added in panel
- [2] New: Svg image button control added in toolbox designer
- [3] Bug fix: diagnostics module
- [4] Bug fix: signal server for arxml multiplex signals

1.567 2023-03-17

- [1] New: Panel controls now can directly call API without associating system variables
- [2] New: UB bits can be set or reset in FlexRay database viewer
- [3] New: mini program supports LIN frame signal manipulation
- [4] Improved: Automation Module renamed to Graphic Program
- [5] New: APIs for system constants traverse
- [6] Improved: LIN RBS Back
- [7] New: FlexRay filter in trace

1.568 2023-03-13

- [1] Graphics Improved: Split header can be defined
- [2] New: min and max column added in graphics
- [3] Graphics improved: Y axis will be invisible if too many signals in list
- [4] Graphics improved: performance improved in cursor selection
- [5] New: min and max value display in Numeric
- [6] New: measurement time display in main window
- [7] New: 29 Win32 APIs added for CAN, LIN and FlexRay database parser
- [8] Crash bug fix in TSMaster for overall stability
- [9] Overall performance improved
- [10] Bug fixes in Automation Module
- [11] Benchmark added: project load time

1.569 2023-03-08

- [1] Performance Improved: CPU load reduction during simulation
- [2] New Video: Python call mini program API
- [3] New Video: TOSUN UDS Module
- [4] New Devices supported: GW2116, TC2115, MP1013
- [5] New: CAN RBS supports setting message direction in UI
- [6] New: Toolbox controls demo added: group box, panel, radio group, shape, scroll bar

1.570 2023-03-07

- [1] Improved: CAN RBS supports multiple CRC and Rolling Counter
- [2] Improved: FlexRay RBS supports multiple CRC and Rolling Counter
- [3] Improved: online replay supports direct ASC file replaying
- [4] New: start and stop buttons added to TSMaster taskbar icon
- [5] New: progress bar added to TSMaster taskbar icon
- [6] Bug fix: enable and disable graphics may cause crash during measurement
- [7] Bug fix: dll function initialize_lib_tsmaster_with_project
- [8] Bug fixes for diagnostics
- [9] Bug fixes for calibration

1.571 2023-02-27

- [1] New: demo for combobox, listbox and static text in toolbox
- [2] New: UB bit auto set and clear implemented in FlexRay RBS
- [3] Bug fix: FlexRay RBS engine in TSMaster.dll
- [4] New: APIs for panel selector list read and write
- [5] Bug fix: mp API blf_to_asc and asc_to_blf
- [6] New APIs: tscom_flexray_set_signal_value_in_raw_frame, tscom_flexray_get_signal_value_in_raw_frame and tscom_flexray_get_signal_definition

1.572 2023-02-02

- [1] New: Video added for toolbox builder design procedure
- [2] New: mini program C API can now be called by python directly
- [3] New: Syntax prompt in python can be automatically generated when mini program API is built
- [4] New: Usage demos for form, label, button, edit are available in toolbox
- [5] New: FlexRay database APIs available in Win32
- [6] New: FlexRay database APIs available in In Process COM
- [7] New: FlexRay database APIs available in Out of Process COM
- [8] New: fibex parser demo implemented in python

[9] Bug fix: blf logging of FlexRay encounter crash in some cases

[10] Improved: Calibration module

1.573 2023-01-15

[1] New: Python environment settings for TSMaster

[2] New: TSMaster package installer for python

[3] New: Copy, cut and paste components supported in TS UI designer

[4] New: You can now use VSCode to define your own TSMaster

[5] Same python code can run in TSMaster or standalone

[6] In standalone mode, all TSMaster windows become your tools

[7] Debugging and stepping is possible using VSCode

[8] New: TS UI Designer can develop and start standalone app now

[9] Improved: program name conflict of mini program is prevented

[10] Bug fix: blf logging for FlexRay events

[11] New: prevent popping up crash dialog option added

1.574 2022-12-30

[1] New: Support multiple logging modules

[2] Update: support create new file after filesize more than setting filesize.

[3] Bug fix: Resolve bugs that cause UI crash of diagnostic module.

[4] Update: FR scheduling table can be configured at the same time in RBS UI and hardware configuration UI.

[5] Update: support Select/Deselect All FR frames of ECU node.

1.575 2022-12-20

[1] New: Toolbox designer released, everyone can design TSMaster professional UI using Python !

[2] New: IDE for python GUI design

[3] New: auto code generation for python GUI

[4] New: Console for integrated python, which can access TSMaster internals in real-time

[5] New: Signal Comparer Module added

[6] New API: API added for Adding signals to Signal Comparer Module

[7] New: Add CAN/CANFD Sequence Transmit Module

[8] Multi-language for Diagnostic Module is completed

[9] Update Calibration Module, support download multi-hex files

1.576 2022-12-06

[1] New: 27 database APIs added for CAN, LIN and FlexRay signal traverse

[2] New Demo: Database signal traverse

[3] New: 2 Test APIs added for signal jump and unchange behaviors

[4] New: Automation Module now supports FlexRay Simulation

1.577 2022-12-01

[1] New: FlexRay Error event, cycle start event and status event supported in trace

[2] New: FlexRay Remaining Bus Simulation feature added

[3] New: Panel now supports FlexRay RBS frame triggering

[4] New: 21 FlexRay RBS APIs added for mini programs and automation modules

[5] New: 10 Test APIs added for signal edge, monotony and signal follow checking

1.578 2022-11-20

[1] C Code editor import supports merge mode

[2] Last value can be displayed in variable list in automation module after run

[3] Improved: Automation module saving and signal selection

[4] C Code Editor supports Flexray rx, tx and preTx events

[5] 13 APIs added for FlexRay

[6] FlexRay database symbols supported

[7] Improved: solution for compilation on computers with Chinese user name

1.579 2022-11-08

[1] log info can show module, function, line number and crash detection

[2] New settings in C code editor

[3] CAN trace can import and export blf and asc

- [4] bus replay can now directly replay asc files
- [5] drag asc file into TSMaster supported
- [6] TS virtual channel now supports FlexRay and Ethernet
- [7] application channel can be renamed
- [8] New: FlexRay Database window
- [9] New: FlexRay Trace Window
- [10] New: CANoe Controller
- [11] New: CANoe Controller APIs
- [12] New: automation module action group supports repeat count as variable
- [13] New: FlexRay Transmit Window
- [14] New: FlexRay signal graphics support
- [15] New: FlexRay signal Numeric support
- [16] New: Video tutorials added for CANoe controller
- [17] New: FlexRay signal meter support
- [18] New: FlexRay panel display support

1.580 2022-09-19

- [1] New: Symbol Mapping now supports expression mapping
- [2] New API: API added for symbol mapping automation
- [3] New: Advanced symbol mapping demo project
- [4] New: Signal generators in CAN Transmit, LIN Transmit and STIM window can now link system variable
- [5] New Window: Constant manager added
- [6] New: User constants feature supported
- [7] New API: get_constant_double to retrieve constant value in code
- [8] Improved: Calibration XCP transport layer now supports CAN FD
- [9] New Video: Advanced symbol mapping

1.581 2022-09-07

- [1] New: System variables can now be logged and replayed
- [2] New: Date time supported in CAN Trace
- [3] New API: system var logging property can be set and get
- [4] New API: get_system_var_address for direct system var memory access
- [5] New API: atomic operations
- [6] Bug fix: graphics splitted view legend refresh

1.582 2022-08-19

- [1] New: Post processing for offline replay
- [2] New: Matlab Controller toolbox re-added
- [3] New: Auto SIL & HIL code generation and mini program integration
- [4] New: Symbol mapping direction can be set
- [5] New: Auto code and panel generation in system variable manager
- [6] Improved: Rolling counter limit supported in CAN RBS
- [7] New video tutorial: Automation module reuse
- [8] New video tutorial: Automation module verdict
- [9] New video tutorial: Checksum and rolling counter native support in CAN RBS
- [10] New video tutorial: J1939 multi-frame transmission and reception
- [11] New video tutorial: Matlab controller SIL and HIL
- [12] New video tutorial: offline replay post processing
- [13] Bug fix: local var not accessible in automation module
- [14] Bug fix: popup window size not correct under high dpi

1.583 2022-08-10

- [1] Improved: Undo and Redo supported in Panel
- [2] Improved: Undo and Redo supported in Automation Module
- [3] Improved: CTS and RTS supported in J1939 TP
- [4] New: Persistent select mode added in Graphics with single select activated

- [5] New: Native support for Rolling counter simulation in CAN RBS
- [6] New: Native support for Checksum simulation in CAN RBS
- [7] New API: CAN RBS function for signal type settings: normal, RC and CRC
- [8] New: UI hang detection
- [9] Improved: CAN RBS UI performance
- [10] New: sub group supported in toolbar sub menu items

1.584 2022-08-01

- [1] Improved: Graphics display performance
- [2] Improved: project path with Chinese characters supported
- [3] New: J1939 Node Configuration dialog
- [4] New: J1939 API for J1939 protocol related operation and multi-packet transmission
- [5] CAN RBS supports J1939 BAM and CMTD PDU transmission and reception
- [6] Measurement windows all support J1939 multi frame signals
- [7] mini program supports J1939 signal get and set through CAN RBS and signal server
- [8] New: Signal checker APIs for automated testing
- [9] Bug fix: calibration log replay with signal name xxx.0.xxx supported
- [10] Bug fix: CAN Trace copy message as text
- [11] Deleted: Internal python x64 distribution

1.585 2022-07-21

- [1] New: send raw message supported in Automation module
- [2] New API: clear measurement form for Graphics, Trace, ...
- [3] Improved: Numeric now supports bar display
- [4] Bug fix: more than one channel assign same database causes signal server display wrong value in graphics, numeric, panels and so on
- [5] Bug fix: panel relation chart cannot distinguish var types of x and y axis

1.586 2022-07-19

- [1] New: error line can be located in C editor with editing features enabled

- [2] New: search all and replace all supported in C editor
- [3] New API: CAN RBS fault injection API added with message lost and signal alter features
- [4] New API: static PDU container location in arxml can now be adjusted only in cycle 0
- [5] Bug fix: modal form has invalid caption in tabbed MDI mode
- [6] Bug fix: system variables associated in panel and numeric not update after application connected more than one time

1.587 2022-07-13

- [1] New API: CAN RBS now supports setting direction of node messages, this can be very useful when there is only one target ECU in dbc or arxml file
- [2] New: up and down indicator for Calibration variables
- [3] New: baseline settings added for Calibration variables
- [4] New: Address mapping editor in calibration memory segment page
- [5] Improved: simple signal group transmission supported in arxml file
- [6] Improved: on message event in C editor now supports code template
- [7] Bug fix: importing a2l with duplicated group name causes UI crash
- [8] Bug fix: importing a2l with duplicated measurement name causes crash
- [9] Bug fix: repeat count in CAN transmit and STIM not functional
- [10] Bug fix: automation module copied group saving causes action overlap
- [11] Bug fix: disconnect app cause calibration parameter curve crash
- [12] Bug fix: transmit load configuration has no effects

1.588 2022-07-04

- [1] New: Auto run property introduced for mini program library without source code
- [2] New: instant API picker supported in automation module
- [3] Bug fix: CAN FD message with DLC > 8 parsed as DLC = 8 in CAN RBS
- [4] Bug fix: adding panel control may cause AV after other panel controls deleted

1.589 2022-06-27

- [1] New: panel get and set apis for UI automation
- [2] New Demo: Tetris game build with panel

[3] New: virtual CAN and LIN messages can now be injected into TSMaster system

[4] New: STIM signals can now be automated using mini program API

[5] New: Automation module can now be automated using mini program API

1.590 2022-06-17

[1] New: Open project dialog added

[2] New: A2L synchronization module

[3] New: async APIs added for system var write

[4] Bug fix: action lost sometimes in automation module after saving

[5] Bug fix: 4 API arguments updated, please update your client code if necessary:
can_rbs_activate_network_by_name, can_rbs_activate_node_by_name,
can_rbs_activate_message_by_name, can_rbs_set_message_cycle_by_name

1.591 2022-06-13

[1] New: App Builder feature added

[2] New: Custom controls supports icon change, import and export

[3] New: Log file path is registered as system variable for automation

[4] New: Project Settings window added, supporting project naming

[5] New: Favorites added in automation module

[6] More shortcuts available in automation module

[7] Improved: tx and rx messages in arxml are now fully parsed

[8] bug fix: extended FD frame from arxml read as standard FD frame

1.592 2022-05-31

[1] New: Custom controls supported in panel

[2] New: copy screenshot to clipboard

[3] Improved: MATRIX_DIM supports single number configuration in calibration a2l file

[4] Bug fix: CAN hardware channel count overwritten by LIN database window

[5] Bug fix: Panel shortcut key not functional

1.593 2022-05-23

[1] New: Encrypted Publishing Toolbox

- [2] New: UDS Code Generator Toolbox
- [3] New Demo: UDS Code Generate with dynamic parameters
- [4] New: App manager
- [5] Improved: Multiple Channel filters supported in CAN Trace
- [6] Improved: Panel layout for scaled and fit are only active when panel is running
- [7] Improved: Export graphics data between measurement cursors
- [8] Improved: Mini program step function now supports timer min step = 0.5 ms

1.594 2022-05-07

- [1] Diagnostics New: Diagnostic debugging module
- [2] Diagnostics New: UDS Diagnostic analysis and display module. Parse the diagnostic data parameters and display them in the tree structure
- [3] The introduction of system variables as the diagnostic parameter type increases the flexibility of the diagnostic module and facilitates data interaction with other modules in tsmaster
- [4] Diagnostics New: executable file parsing and downloading module
- [5] Diagnostics New: UDS test group and test case management
- [6] Diagnostics Update: Automatic diagnostic flow module. Diagnostic flow such as flash bootloader can be completed through this GUI module without coding.
- [7] New: MP API added for system var properties setting and rbs
- [8] Improved: in online replay, one source channel can be mapped to multiple destination channels by inserting comma between channel numbers
- [9] Improved: Hex display format supported in Numeric

1.595 2022-04-26

- [1] New: Symbol mapping now supports bidirectional mapping with RBS support
- [2] New: Videos of TSMaster real-time HIL simulation for automotive ECU development
- [3] New: Application Event order for advanced users
- [4] New: CPU core tuning for advanced users
- [5] Improved: system var supports pointer type argument in graphical language
- [6] Graphics bug fixes

1.596 2022-04-18

- [1] New: Build RBS for specific node (HIL mode)
- [2] New: AN0006 (Use automation module and graphical language) is published
- [3] Bug fix: Video replay window cannot be opened

1.597 2022-04-16

- [1] New: CAN database signal based communication matrix viewer
- [2] New: CAN database message based communication matrix viewer
- [3] Improved: Calibration reconnect feature implemented
- [4] Bug fix: CAN database configuration corrupt with Chinese characters in file path

1.598 2022-04-12

- [1] New: report generator added in automation module
- [2] New: constant picker added in c code editor and automation module
- [3] New: copy paste of single action supported in automation module
- [4] New: graphical language sub system tutorial video
- [5] New: automation module navigation feature
- [6] Improved: graphics drag drop insertion
- [7] Improved: Calibration Parameter Curve now supports value table
- [8] Note: to modify multiply selected items, press enter after selection, value picker list will be popued

1.599 2022-04-06

- [1] New: Graphics now supports split views
- [2] Improved: Graphical language non-decision actions omits direction

1.600 2022-03-30

- [1] New: Automation module replay feature
- [2] panel switch component completely rewritten with more color options
- [3] Improved: software performance enhanced in tabbed MDI mode
- [4] Improved: lower CPU consumption with large window number
- [5] Bug fix: switch from tabbed MDI to normal form cause caption display incorrect

1.601 2022-03-26

- [1] Graphical programming language supports sub system and module link
- [2] Graphical programming language supports step into (F7)
- [3] Bug fix: Text in Panel property can selected by cursor drag
- [4] Bug fix: Panel menu display incorrectly in DPI <> 96
- [5] Bug fix: Panel scrollbox display no scrollbars in DPI <> 96

1.602 2022-03-19

- [1] New: change title is now easier on right-top area of each form
- [2] New: set top most window supported
- [3] New: Real-time comments can be added in calibration log files in offline mode
- [4] Improved: Calibration ECU name can be changed
- [5] Improved: write_realtime_comment API now supports blf created by start_log

1.603 2022-03-15

- [1] New: Memory consumption and load time added in settings
- [2] New: Video for graphical programming language debugging
- [3] New: Printing supported in automation module
- [4] New: Filter added for online replay
- [5] New: Filter now supports channel in CAN Trace
- [6] Improved: UI of calibration

1.604 2022-03-10

- [1] New: Automation module arbitrary expression supported
- [2] New: Diagnostics API

1.605 2022-03-07

- [1] New: Automation module logging
- [2] Improved: Calibration curve display

1.606 2022-03-04

- [1] New: Set home and restore home supported in graphics
- [2] Bug fix: crash in calibration measurement list edit with filter

[3] Improved: global font in software

[4] Fixed: database and log file path corrected in demo scripts

1.607 2022-03-02

[1] New: play sound API

[2] New: Goto and From supported in Graphical language

[3] API bug fixes

1.608 2022-02-28

[1] API bug fix

1.609 2022-02-25

[1] New: Variants supported in Automation Module

[2] Improved: row height of graphics becomes smaller to display more signals

[3] Modified: wheel scroll does not pause graphics from now on

1.610 2022-02-22

[1] New: Automation Module released

[2] New: Graphical programming language released

[3] New: software global font can be changed in settings

[4] Improved: Software font looks more clear in machines with DPI <> 96

1.611 2022-02-18

[1] New: Use parsed database file to improve project load speed by 100x faster especially with super large dbc and arxml files

[2] New: Option to copy database file to local project folder (default not copy)

[3] New: Options to control numeric and 3D visibility in calibration curve

[4] New: Options to prevent RBS from being activated by external modules such as panel

[5] Improved: ECU name existence can be controlled also in mat and mdf file

[6] Bug fix: Graphics import data cause hang

[7] Bug fix: Calibration ECU memory and cache insistency

1.612 2022-02-14

[1] New: option to add ECU name prefix in system vars generated by calibration

[2] bug fix: graphics displays value table and units during calibration replay

[3] minor bug fixes and improvements

1.613 2022-02-08

[1] New: MP variable init value supported

1.614 2022-01-27

[1] Improved: dead-lock of on var change event auto break in C code execution

[2] Improved: Panel UI load and unload performance increased

1.615 2022-01-26

[1] New: project directory being used prompt supported

[2] Bug fix: db changes cause hidden panel crash

[3] New: auto generation of C code on compile for Git version compare

1.616 2022-01-22

[1] new file type: PT7z (password protected T7z)

[2] TSMaster project directory icon supported

1.617 2022-01-21

[1] directory based project management instead of T7z file

[2] Note: all relative paths reference to project directory

[3] Note: temporary directory can be dangerous, please create new directory for each project

[4] Note: T7z file is now only used to import, export and project exchange

[5] Note: Git can now be used on project directory

1.618 2022-01-17

[1] Single signal mode supported in graphics

[2] horizontal cursor and vertical cursor supported in graphics

[3] + - * / buttons supported in Parameter curve

[4] minor bug fixes in graphics

1.619 2022-01-13

[1] New: signal server API supported in mini program

[2] Improved: Numeric now supports replay

1.620 2022-01-11

[1] New: Arxml Multiplex signal display supported in Graphics, Numeric, Meter, Panel

[2] New: Factor and Offset supported in symbol mapping

[3] Improved: Panel high display quality

1.621 2022-01-03

[1] Mini program variable change event can associate any system variable

[2] Module Auto start priority can be set in software settings

1.622 2021-12-26

[1] Kernel improvement: load database, startup performance, etc.

[2] graphics bug fix: cursor time incorrect

[3] API Improvement: API can be initialized multiple times for LabVIEW

1.623 2021-12-21

[1] New: system variables can now be created or deleted by API

[2] Improved: C editor system function names sorted

[3] Bug fix: unsupported arxml multiplexed signals deleted in C editor

[4] Bug fix: delete form in measurement setup causes order incorrect

1.624 2021-12-14

[1] Improved: CAN Trace redesigned (arxml pdu supported)

[2] Improved: CAN Trace sort dialog added

[3] Improved: Project load speed increased

[4] Improved: MDI title bar double click handler

1.625 2021-12-09

[1] New: [Calibration] Array type Measurement Var monitor, log and replay supported

[2] Bug fix: enum display in calibration

[3] Bug fix: factor and offset calc. in graphics

1.626 2021-12-06

[1] New: CAN Trace Sync. token with Graphics supported

[2] New: C Code snippet supports DB signals and RBS signals

[3] Improved: Graphics horz. and vert. adjust

[4] Improved: Graphics new column: Type

[5] Improved: System messages new filter

[6] Improved: C Code editor compiler default supports multi thread

1.627 2021-11-26

[1] New: Get unique computer hw id api added

[2] Improved: TSMaster icon resolution enhanced

[3] Improved: all signals in message can be auto created as pair in symbol mapping

1.628 2021-11-24

[1] Improved: C editor omits duplicated message name while compiling

[2] Bug fix: edit signal in sorted list of graphics

1.629 2021-11-15

[1] New: tutorial videos in youtube when language is not Chinese

[2] New: drag signals from Calibration database and System variable manager to STIM

[3] Improved: UI elements in CAN database, CAN Statistics

[4] Bug fix: default byte order in Calibration

[5] Bug fix: force stop connecting in calibration cause crash

1.630 2021-11-04

[1] New: Add, Dec, Mul, Div supported in STIM

[2] New: Diagnostics

1.631 2021-10-26

[1] New: auto code gen in CAN RBS in popup menu

[2] New: API to set message cycle in CAN RBS

[3] Improved: address in A2L now supports 0xXXXXL (non official)

[4] Improved: mouse wheel on signal Y axis in Graphics: scroll to move, Ctrl + scroll to zoom

[5] Bug fix: get or set vendor detect options API

1.632 2021-10-19

- [1] New: Counter column in CAN Trace
- [2] New: Graphics y axis display option
- [3] Improved: quantization error in trace eliminated
- [4] Improved: RAM and Flash mapping supported in calibration for A2L generated by Vector
- [5] Bug fix: clear bus stat under baud <> 500k
- [6] Bug fix: multiplex editor in graphics

1.633 2021-10-06

- [1] New: Open source CAN tool CANable (candleLight) supported

1.634 2021-10-03

- [1] Improved: High DPI display
- [2] New: compiler parameters in C Editor

1.635 2021-09-29

- [1] New: Test system parameters can now be loaded from ini file
- [2] New: Test system parameters list load and save with csv supported

1.636 2021-09-28

- [1] New: Drag drop from Trace to Graphics, Meter, Numeric, Panel
- [2] New: Drag drop from Calibration database to Graphics, Numeric, Parameter Curve, Panel
- [3] New: Drag drop from CAN Database to Graphics, Numeric, Panel
- [4] New: Drag drop from LIN Database to Graphics, Numeric, Panel
- [5] New: Drag drop from System Var Mgmt to Graphics, Numeric, Panel
- [6] New: Default settings options for STIM, CAN Transmit, CAN Trace, LIN Transmit, LIN Trace, Graphics
- [7] Improved: Calibration log file name supports absolute dir
- [8] Improved: Graphics signal list header popup menu added

1.637 2021-09-25

- [1] Improved: App Host options

[2] Bug fix: calibration new ecu cause crash

1.638 2021-09-22

[1] New: API for ini files read and write

[2] Improved: var on written and on change distinguished in mini program

[3] Improved: Graphics and STIM now highlights deleted signals

[4] Bug fix: calibration CRC DWORD algorithm selected as ARC

[5] Bug fix: calibration hex export with address range > 0xFFFF

1.639 2021-09-16

[1] Improved: stop measurement in each MDI is invoked before app disconnect

[2] Improved: batch modification in graphics

[3] Improved: value table of system variable is now ordered

[4] Improved: Calibration measurement list highlights n.a. signals

[5] Improved: Calibration measurement list filter implemented

[6] Bug fix: set double signal in mini program

[7] Bug fix: keydown enter twice in graphics

[8] Bug fix: prevent graphics eat mouse click after right mouse button down

[9] Bug fix: Non-ISO cannot be selected in TS FD hardware config

[10] Bug fix: value table lost in copied message in CAN FD Transmit window

1.640 2021-09-11

[1] Improved: Graphics now supports undo and redo

[2] Improved: Graphics grid color alpha set to 20%

[3] Improved: Graphics X axis font color same as X axis color

[4] Improved: Graphics Y axis increment max count can be set

[5] Improved: Numeric window now supports offline bus signal replay display

[6] Improved: Calibration error code message display added

[7] New: Prompt user input in MP API

[8] Bug fix: panel drag control over container causes hang

1.641 2021-09-03

- [1] Improved: Graphics supports signal color customization
- [2] Improved: Graphics zoom feature
- [3] Improved: Kill focus on panel input output box
- [4] New: Calibration CRC algorithm: CRC_BYTE, CRC_DWORD, CRC_2_WORD
- [5] Bug fix: Panel enable property
- [6] Bug fix: blf write API without timestamp causes incorrect measurement start time
- [7] Bug fix: first sample point not displayed in graphics

1.642 2021-08-31

- [1] Bug fix: large raw signal modify in CAN FD transmit
- [2] Improved: MDI form borders are easily to drag
- [3] Improved: Graphics signal list width persistence

1.643 2021-08-29

- [1] New: Graphics goto specific time
- [2] New: Sync scroll among all Graphics, based on their "Sync. Token" property
- [3] New: fixed form mode (cannot resize and move)
- [4] Bug fix: format display in calibration

1.644 2021-08-26

- [1] Bug fix: IEEE float 32 and 64 factor and offset not calculated
- [2] New: Graphics signal can switch to vertical line junction style
- [3] Updated: save measurement dialog add delete file option

1.645 2021-08-25

- [1] Bug fix: Panel pasted control drag causes shadow

1.646 2021-08-24

- [1] Bug fix: calibration stop uploading while connecting cause hang
- [2] Bug fix: CCP byte order cannot be changed
- [3] Bug fix: blf BL_OB_TYPE_CAN_FD_MESSAGE_64 parse
- [4] Improved: up and down button affect all selected signals in STIM

[5] Improved: Graphics drag drop multiple signals supported

[6] Improved: Curve drag drop multiple signals supported

1.647 2021-08-18

[1] New: Graphics line style and width options

[2] New: mat file read write api added

[3] Updated: Calibration Curve order list by dragging

[4] Bug fix: some panel configuration lost after one of multiple panels deleted

[5] Bug fix: Trace display LIN 3C 3D signal crash

1.648 2021-08-12

[1] Bug fix: CAN FD transmit window, signal init value is always 0

1.649 2021-08-10

[1] New: CAN Trace set reference under chronological mode

[2] New video: calibration automation

[3] Bug fix: Calibration cache enable option persistent

1.650 2021-08-08

[1] New demo: Convert Blf to signal based CSV

[2] Updated: real-time comment is now shown in Graphics charting area

1.651 2021-08-06

[1] Bug fix: calibration crash when stopping in polling mode

[2] Updated: C Code Editor var and timer highlighted, method popup supported

1.652 2021-08-05

[1] New: udp APIs added in ioip

[2] Bug fix: Panel circular reference problem

[3] Bug fix: Align of child objects in panel is now persistent

[4] Bug fix: channel selection dialog select all and deselect all

1.653 2021-08-04

[1] New: Auto Mapping Code generation for TSMaster SDK

1.654 2021-08-03

[1] New: file and directory API added

[2] New: UDP broadcast API added

1.655 2021-08-02

[1] New: Global rx filter added for all supported CAN interfaces

[2] New: Feature select with project templates

[3] New: video link added: Global Rx filter

[4] Bug fix: Panel start before mini program causes system var link incorrect

1.656 2021-07-30

[1] New: video link added: Mini Program tutorial 2

[2] New: Feature select dialog

[3] New: text file APIs in mini program

1.657 2021-07-29

[1] New: video links added in Calibration window, C Editor, System Var Mgmt, Hardware

[2] New: AN0005 How to transmit LIN frames

[3] Bug fix: CAN trace display mode change causes frame lost

[4] Updated: Calibration A2L group with no parent will be shown in root

1.658 2021-07-27

[1] New: calibration parameters can be inserted to hex

[2] New: TSDB added in out of process server

[3] New: system variables can be read or write by COM server

[4] New: video links added in C Code editor and CAN RBS windows

[5] Improved: TSMaster setup file may not be killed by some virus software

[6] Improved: TC1005 driver performance enhanced

[7] Updated: TSMaster Calibration document AN0001

1.659 2021-07-23

[1] New: Measurement window: Numeric

[2] New: server response API functions added in ioip in mini program

- [3] New: calibration parameters can be imported from hex
- [4] New: buttons on quick customization bar on main ribbon
- [5] Bug fix: panel input-output box initial value is always n.a.
- [6] Bug fix: calibration map batch modification not work

1.660 2021-07-15

- [1] Mini program VC++ project can be generated directly in editor
- [2] MDI now supports colorful window title color

1.661 2021-07-14

- [1] SYSTEM_CONSTANT related compu methods supported in calibration
- [2] TSMaster plugin can also be implemented as mini program library
- [3] Offline calibration mode supported in XCP
- [4] Offline calibration mode supported in CCP

1.662 2021-07-13

- [1] realtime comments now supported in bus log
- [2] APIs added for blf realtime comments
- [3] APIs added for realtime comments in app environment
- [4] Blf_Logging example updated for realtime comments
- [5] Build checksum supported in XCP
- [6] Build checksum supported in CCP

1.663 2021-07-08

- [1] Graphics all columns can now be sorted
- [2] Trace fixed display logic re-written

1.664 2021-07-07

- [1] CAN Trace fixed mode stress test (100000 frames/sec per channel) passed
- [2] Graphic signals support sorting by name
- [3] Bug fix: Calibration large address cause crash (not tested)
- [4] Bug fix: Test system report automation image template generation

[5] Added: post process script supported in Test system report automation

1.665 2021-07-01

[1] CCP hex file can be exported after calibration

[2] Bug fix: bkgd color read crash in panel

[3] Bug fix: TS virtual channel mapping incorrect

[4] XCP hex file can be exported after calibration

[5] XCP hex file can be uploaded and downloaded

1.666 2021-06-26

[1] Released: Mini Program SDK (<https://github.com/TOSUN-Shanghai>)

[2] CCP hex file can be uploaded and downloaded

1.667 2021-06-20

[1] Panel controls now support value table from system variables

[2] Graphics refreshes during offline replay

[3] Bug fix: CAN RBS not run first time when a configuration is loaded

[4] TSMaster T7z file is now compatible with 7zip file format, replace files in project configuration manually is allowed

1.668 2021-06-17

[1] Panel input output box bkgd and border color, font can now be customized

[2] new API: `app.get_tsmaster_version`

[3] value table supported in STIM window

[4] A2L parser now supports "ECU" as module name

[5] Panel bug fix: copy paste controls cause some objects missing

[6] dialog added for unexpected start of CAN RBS engine in panel editor

1.669 2021-06-14

[1] Test system example updated: insert screenshots into word report

[2] Test API added: `retrieve_current_result_folder`

[3] Test API added: `write_result_image`

[4] Test system new example: Excel test system report automation

1.670 2021-06-12

[1] Excel test system report process script demo added

1.671 2021-06-11

[1] Improved: A2L loading speed increased by 10x

[2] Bug fix: TC1011 baud rate UI not refreshed after CAN core switched

[3] Bug fix: custom signal channel read only in graphics

1.672 2021-06-08

[1] New Feature: TCP Server, TCP Client, UDP Server and UDP Client are supported in Mini program API

[2] Bug fix: First value display error in Calibration Curve under Motorola byte order

[3] System variable now supports unit display, value table display, accuracy, format to string, affects display of Calibration signals also

[4] New Ethernet demo added: Generic TCP UDP Communication

1.673 2021-06-04

[1] Bug fix: CCP fixed CAN ID DAQ packet data corrupt

[2] Bug fix: Calibration save log dialog null value cause crash

[3] Signal comments in transmit window are now persistent

[4] Channel display added in database selector popped by C Editor

[5] online replay engine stop criteria changed to > 1000 Tx errors

1.674 2021-06-02

[1] CCP DAQ with fixed CAN identifier and event channel supported

[2] MP API header fixed: excel_get_cell_value

[3] MP API added: get_configuration_file_name

[4] MP API added: get_configuration_file_path

1.675 2021-05-31

[1] New Feature: Window layout control

[2] Bug fix: dbc cycle parse

1.676 2021-05-29

[1] Bug fix: CAN RBS load configuration not refresh UI

[2] Bug fix: display of AXIS_PTS in calibration database window

1.677 2021-05-26

[1] Bug fix: crash on connection to some HW

1.678 2021-05-25

[1] CAN transmit window signal step settings are now persistent

[2] mini program library dependency is now persistent

[3] CAN RBS node selection is now persistent

[4] CCP commands SET_S_STATUS and SELECT_CAL_PAGE supported

[5] dbc parse speed 100x faster

1.679 2021-05-21

[1] Timer accuracy optimized

1.680 2021-05-20

[1] CAN RBS topology chart added

1.681 2021-05-14

[1] max LIN frames in schedule table set to 150

1.682 2021-05-13

[1] Blf logging API demo configuration added

[2] Bug fix for extended frame id in asc conversion

1.683 2021-05-10

[1] Each MDI window's z-order is now remembered

[2] Blf file API added

1.684 2021-05-05

[1] mini program library dependencies configuration supported

1.685 2021-05-02

[1] TSMaster plugin manager released

1.686 2021-04-28

1.687 2021-04-23

[1] Signal relation graphics added in panel

1.688 2021-04-22

[1] TSMaster software architecture completely redesigned

1.689 2021-04-16

1.690 2021-04-13

[1] minor bug fixes

1.691 2021-04-09

[1] carlibration par file import supported

[2] fifo module supported in COM API

[3] graphics cursor zoom feature supported (default right cursor)

[4] bug fix: bus logger generates additional frame

1.692 2021-04-02

[1] improved support for arxml import

[2] multiple databases in arxml can be extracted

1.693 2021-04-01

[1] editing of measurement data comment supported in graphics

1.694 2021-03-31

[1] CCP feature release

[2] measurement cursor in graphics: left click to position, right click to cancel position

1.695 2021-03-26

[1] TSMaster and API overall performance improvement

[2] feature added: dbc, ldf, excel test case auto reload after edit

[3] default upload all calibration variables on connection

[4] calibration comments can be displayed when cursor hovers mat file

[5] par file export in calibration

[6] bug fix: chinese char in transmit and graphics window not display

1.696 2021-03-22

[1] TSMaster file preview shell components supported

[2] shortcuts (F9, F10 and ESC) supported in graphics

[3] string type supported in Text, input-output box and selector in panel

[4] real-time comment supported during measurement (ALT + C)

[5] graphics integral time display supported and made default

[6] calibration database list display mode added

1.697 2021-03-17

[1] bug fix: caption rename of curve form

[2] wildcard search supported in calibration, CAN DB, LIN DB, system variables and symbol mapping

[3] comments dialog support for calibration measurement files

[4] parameter curve undo redo supported

[5] parameter curve copy paste supported

[6] STIM multiple change supported

[7] STIM undo redo supported

1.698 2021-03-15

[1] all calibration scalar signals are now supported in panel

[2] curve and map data multi-select-modify supported

[3] curve and map data interpolation supported

1.699 2021-03-12

[1] Bug fix for Calibration data timestamp display

[2] Bug fix for Map COLUMN_DIR display

[3] Bug fix for graphics system variable display

1.700 2021-03-10

[1] Calibration log can be exported to MATLAB replay model

[2] TSMaster LIN API updated

1.701 2021-03-09

[1] Curves and Maps supported in calibration

[2] MF4, MDF and DAT log files import supported in Calibration (2.00, 2.10, 2.14, 3.00, 3.10, 3.20, 3.30, 4.00, 4.10, 4.11, 4.20) from CANape and INCA

1.702 2021-03-02

[1] MDF log files export supported in Calibration (2.00, 2.10, 2.14, 3.00, 3.10, 3.20, 3.30, 4.00, 4.10, 4.11, 4.20) for CANape and INCA

[2] A2L parser crash problem fixed

1.703 2021-02-25

[1] API for UI automation added

1.704 2021-02-24

[1] COM_AXIS supported in XCP Curve

[2] COMPU_VTAB, COMPU_TAB, FFORMULA supported in XCP

[3] Graphics will show never-appear signals as n.a.

[4] fixed: empty blf file replay cause TSMaster never connect

[5] time axis also shows realtime value in Graphics

1.705 2021-02-22

[1] STIM custom signal file import and export supported

[2] Excel Test module released

1.706 2021-02-20

[1] PEAK CAN FD device supported

[2] Bug fix: FCOEFFS_LINEAR parse in XCP

1.707 2021-02-10

[1] Video step method in AN0004 document updated

[2] shortcuts supported in video replay

[3] TSMaster features document updated

[4] LIN diagnostics API Demos added

[5] github link added: <https://github.com/TOSUN-Shanghai/TSMaster>

1.708 2021-02-06

[1] video step supported in video replay

[2] C# API updated

1.709 2021-02-05

[1] writeable measurement signal supported in XCP

- [2] array parameter supported in XCP
- [3] Axis parameter supported in XCP
- [4] mini program library API exported
- [5] LIN diagnostics API exported
- [6] curve with fixed axis supported in XCP
- [7] video replay with blf feature supported
- [8] application note AN0004 added: video replay

1.710 2021-01-25

- [1] motorola supported in XCP module
- [2] bug fix for TC10xx USB transfer protocol
- [3] CAN multiplexor signal and multiplexed signal supported in Graphics

1.711 2021-01-24

- [1] CAN multiplexor signal and multiplexed signal supported in trace window
- [2] CAN multiplexor signal and multiplexed signal supported in transmit window
- [3] CAN signal sorted by position in trace window
- [4] CAN multiplexor signal and multiplexed signal supported in CAN RBS
- [5] CAN multiplexor signal and multiplexed signal supported in APIs

1.712 2021-01-21

- [1] Solved: panel align persistency
- [2] panel objects hierarchy is now displayed in tree
- [3] variables and timers documents now available in c editor
- [4] comments translated in c editor
- [5] group name of radio button persistency added in panel
- [6] alignment of signal text in CAN trace
- [7] xcp manual configuration without a2l supported
- [8] selector in panel will not display "=" in the future
- [9] special thanks list updated

[10] C Mini program now supports C++ 11 standard

1.713 2021-01-18

[1] system variable logging in calibration done

[2] system variable replay in calibration done

[3] symbol mapping from CAN signal to system variable done

[4] legacy TC1005 firmware supported

[5] Calibration module translated

[6] application note feature added, which can be found on top-right area of TSMaster standard form, for example, Calibration form

[7] application note for Calibration added: AN0001: How to use TOSUN Calibration XCP controller

[8] application note for Calibration added: AN0002: Seed and Key algorithm implementation in XCP controller

[9] application note for Calibration added: AN0003: TOSUN XCP controller mat log file format

1.714 2021-01-09

[1] Scalar signals in XCP module now available

1.715 2021-01-04

[1] TSMaster Pre-Tx API added in C version

1.716 2020-12-30

[1] TSMaster C API released, which can be found in "bin\Data\SDK\examples\C\"

1.717 2020-12-24

[1] Mini program library demo project added: Simple Gateway Implementation

[2] The running state of Mini program library can be triggered by system variable

[3] The running state of calibration can be triggered by system variable

[4] The value of Internal System variable can be modified directly in "System Variable Manager"

[5] Add dialog support for blf logging file save operation

[6] Default point count of Graphics set to 100000, can be adjusted

[7] Stimulation feature added

1.718 2020-11-28

[1] Panel LED now has Threshold property, under threshold mode, threshold high and low properties are active, otherwise, single ONValue is active

[2] F5 and F6 shortcuts are added for starting and stopping application

1.719 2020-11-27

[1] TS virtual channel com server removed

[2] asc file time supported

1.720 2020-11-19

[1] bug fix: can error frame read in blf

[2] c code can be generated from tx and trace window

1.721 2020-11-18

[1] Bug fix: CAN trace add filter

[2] Bug fix: panel graphics and pie edit signal index out of bounds

[3] Panel enable property added

[4] C# SDK updated

[5] LabVIEW SDK added

[6] C# docs updated

[7] Bug fix: graphics signal selection display in y axis

1.722 2020-11-16

[1] panel select enhanced: normal select and invert select

[2] precision property supported in panel text

[3] classification of auto start modules

1.723 2020-11-15

[1] system lang auto detect supported

1.724 2020-11-14

[1] translated: all dynamic languages

[2] bug fix: j1939 dbc cause can rbs crash

- [3] panel ui refresh rate can be set
- [4] input output box in panel has label width property
- [5] switch in panel has label property
- [6] path button in panel has label property

1.725 2020-11-13

- [1] printf supported in mini program
- [2] translated: channel selection
- [3] translated: check update
- [4] translated: documents
- [5] translated: graphics
- [6] translated: LIN database
- [7] translated: LIN RBS
- [8] translated: LIN trace
- [9] translated: LIN transmit
- [10] bug fix: panel display with more than 2
- [11] translated: logging
- [12] translated: matlab controller
- [13] translated: measurement data filter
- [14] translated: measurement setup
- [15] translated: meter
- [16] translated: mini program library
- [17] translated: panel
- [18] translated: playback
- [19] translated: settings
- [20] translated: symbol mapping
- [21] translated: system variable mgmt

[22] translated: system messages

[23] panel width and height properties now supported

1.726 2020-11-10

[1] ICS VCAN3 channel 2 MSCAN supported

[2] translated: test system

[3] translated: hardware configuration

[4] translated: ribbon

[5] translated: automotive converter

[6] translated: C2Stateflow

[7] translated: CAN / CAN FD Trace

[8] translated: CAN RBS

[9] translated: bus statistics

[10] translated: C Code editor

[11] translated: channel mapping

[12] translated: vendor selection

[13] panel: press esc to select parent control

[14] splash screen not top most

[15] panel: control name can be hidden by option

1.727 2020-10-31

[1] translated: form manager

[2] bug fix: PEAK receive interface

1.728 2020-10-30

[1] Object list added in Panel

[2] Arbitrary python scripts can be executed sync or async in mini program and TSMaster APIs

[3] max single log file size can be controlled in logging window

[4] multi-language translated: CAN Database

1.729 2020-10-26

- [1] Panel label display format supported
- [2] Panel Graphics bkgd color and transparent properties added
- [3] Panel Units can be displayed in graphics
- [4] Panel input output box initial value corresponds to RBS value

1.730 2020-10-22

- [1] Single J1939 message can be transmit in CAN Transmit window
- [2] Bug fix: Trace not refreshed after J1939 dbc channel switch

1.731 2020-10-20

- [1] dbc to c source default output non-floating points
- [2] output encoding in automotive file converter can be switched between ANSI and utf-8
- [3] API bug fix: show single form by API should hide splash
- [4] API bug fix: load TSMaster.dll should not change current dir
- [5] J1939 single frame can now be displayed in trace with signals expanded

1.732 2020-10-19

- [1] Enumerate Hardware API added

1.733 2020-10-16

- [1] ARXML AutoSAR format supported
- [2] Automotive file converter added in Tools page, supported 10 format:

dbc

arxml

xlsx

xls

dbf

yaml

sym

csv

json

fibex

[3] dbc to C source code feature added in Automotive file converter

1.734 2020-10-13

[1] minor bug fixes in TSMaster API

1.735 2020-10-12

[1] Application Shortcuts supported: ctrl+O, ctrl+N, ctrl+S, ctrl+TAB, ctrl+shift+TAB

[2] hex display on/off will immediately trigger UI refresh

[3] Trace expanded nodes will be remembered

[4] Shortcuts list collected in project overview

[5] life time free components listed in About window

1.736 2020-10-10

[1] Panel section completed in help manual

[2] Project details page in backstage added

1.737 2020-10-09

[1] TC1005 and IntrepidCS now support CAN Transmit API Sync. operation

[2] Help Manual updated, RBS and C Code Editor chapters completed

1.738 2020-10-08

[1] CAN RBS (Remaining Bus Simulation) API added in automation

[2] example added: RBSInExe.py, control RBS in python

[3] example added: CAN Remaining Bus Simulation Scripting.T7z

1.739 2020-10-07

[1] CAN RBS API added in mini program, TSMaster.dll

[2] CAN online replay API added in mini program

1.740 2020-10-06

[1] Test system demo added

1.741 2020-10-05

[1] CAN database message layout chart added

[2] Demo added: "how to use preTx event to implement checksum and rolling counter gracefully"

[3] Measurement window "auto start" feature can now be quickly checked on right-top of window area

[4] Panel controls can now be copied, pasted, moved, aligned, arranged by multiple selection

[5] Title text added in LED control in panel

1.742 2020-10-01

[1] UI: Application button added

[2] Credits added

1.743 2020-09-29

[1] C Sharp SDK samples added

1.744 2020-09-28

[1] Panel UI completed, user can associate signals and monitor, manipulate signals

1.745 2020-09-23

[1] Graphics now supports y axis auto scaling

[2] PEAK auto reset on bus off feature added

[3] Graphics now supports y axis mouse wheel zooming

1.746 2020-09-18

[1] Example projects added, which can be found in "bin\Data\Demo\Projects"

1.747 2020-09-14

[1] TSMaster beta version update supported

[2] TSMaster can be installed in user mode

1.748 2020-09-03

[1] Documents added.

1.749 2020-08-31

[1] System variable feature supported. System variables can be accessed in mini program, graphics and variable management window.

1.750 2020-08-15

[1] Measurement Setup window implemented.

[2] Measurement filter window implemented.

1.751 2020-08-03

[1] new project configuration now supports template

[2] Graphics now supports split view and value table in Y axis

[3] IntrepidCS HSCAN devices supported:

NEODEVICE_BLUE

NEODEVICE_ECU_AVB

NEODEVICE_DW_VCAN

NEODEVICE_RADGIGALOG

NEODEVICE_FIRE

NEODEVICE_VCAN3

NEODEVICE_RED

NEODEVICE_ECU

NEODEVICE_OBD2_PRO

NEODEVICE_PLASMA

NEODEVICE_ION

NEODEVICE_VCAN44

NEODEVICE_VCAN42

NEODEVICE_FIRE2

NEODEVICE_RADGALAXY

NEODEVICE_RADSTAR2

NEODEVICE_VIVIDCAN

NEODEVICE_OBD2_SIM

1.752 2020-07-27

[1] ZLG CAN devices supported:

USBCAN-8E-U

USBCAN-4E-U

USBCAN-2E-U

USBCAN-E-U

USBCAN-I

USBCAN-II

USBCAN2A

USBCAN-E-mini

1.753 2020-07-21

[1] Kvaser CAN devices supported.

1.754 2020-07-18

[1] PEAK USB CAN devices supported.

1.755 2020-07-13

[1] Automation API for online replay implemented.

1.756 2020-07-08

[1] Online replay feature added, which is capable of replaying multiple log files according to their own replay settings.

1.757 2020-06-23

[1] TSMaster in-process server "comTSMaster.dll" is functional. Please check the demo "TestCOMTSMaster.py" under "bin\Data\Demo\Automation\InProcess\".

[2] TSMaster out-of-process server "TSMaster.exe" is functional. Please check the demo "TestTSMaster.py" under "bin\Data\Demo\Automation\OutOfProcess\".

1.758 2020-06-10

[1] Test system feature added, which enables the user to create own test system to perform test cases automatically and creating test reports freely.

[2] Mini program library feature added, mini program can now be used as a library by other mini programs. This is useful for test cases and automation scripts design with test plugin support.

1.759 2020-05-23

[1] CAN and LIN signals can be added to Graphics and Meter windows from CAN or LIN Trace windows by popup menu.

1.760 2020-05-20

[1] Matlab automation window added, which enables the user:

[2] quick find variables in base workspace, plot variables

[3] convert existing C code to pure stateflow, to meet the requirements of stateflow programming guide, the C code should be implemented only using “if” and “else”, other grammars of C code are not supported.

[4] quick add subsystems and charts using known inputs and outputs

1.761 2020-05-18

[1] Magnetic form feature added.

1.762 2020-05-15

[1] “CAN Transmit Window” and “CAN Trace Window” are deleted permanently in future release, use “CAN / CAN FD Transmit Window” and “CAN / CAN FD Trace Window” instead.

1.763 2020-05-12

[1] Firmware version and date of TS USB device are now displayed in TS Channel Mapping window.

1.764 2020-05-11

[1] CAN FD error frame info display of XL devices.

[2] Comment display added under each meter object.

1.765 2020-05-08

[1] CAN databases supported in TS Mini Program.

1.766 2020-05-03

[1] TS Mini Program with C code support released, which can be found in “Simulation” ribbon tab.

1.767 2020-04-04

[1] Turbo mode is added into hardware settings tab. Checking this option will minimize all hardware channel latencies at the expense of consuming more CPU usage. Users who concern very much for hardware performance are recommended to check this option.

1.768 2020-03-26

[1] CAN RBS feature added

1.769 2020-03-24

[1] Max CAN, LIN channel count set to 32

2 Bug Fixes

2021-01-15

[1] Bug fix: libTSH transmit frames lost issue

2020-12-03

[1] Bug fix: trace filter for J1939 id

[2] Bug fix: panel selector crash after signal selected

2020-11-21

[1] bug fix: RBS UI refresh

[2] bug fix: Graphics signal value table display

2020-11-17

[1] Bug fix: J1939 PGN display in Trace

[2] Bug fix: read fd error frame in blf

[3] Bug fix: write log file with TC1005

[4] Bug fix: CAN database filter with msg identifier

[5] Bug fix: liblog.dll not found, please use latest TSMaster.dll in API

2020-10-07

[1] offline replay range playback incorrect behavior

[2] graphics y axis scaling incorrect behavior

2020-10-06

[1] extended frame conversion from asc to blf

2020-09-24

[1] Panel crash on startup fixed

[2] Replay hang when frame count > 100000

[3] bug fix for dbc signal calculation with factor < 0

2020-09-16

Transmit and display problems of signals with mixed value table and factor and offset values.

2020-09-01

ZLG baudrate config now supports devices except "E" series.

2020-05-15

Key trigger of CAN message transmission is not functional.

Ordering of CAN message transmission column is now persistent.

Visibility of CAN message transmission column is now persistent.

2020-05-12

Classical CAN transmission in XL may cause crash.

CAN and LIN databases channel selection treeview display additional "+" button when all child nodes are removed.

Physical step CAN transmit cannot be modified.

2020-05-11

Display error of TOSUN icon in 150% text size mode in win10.

2020-03-24

LIN message "Active" property cannot be modified after deploy.